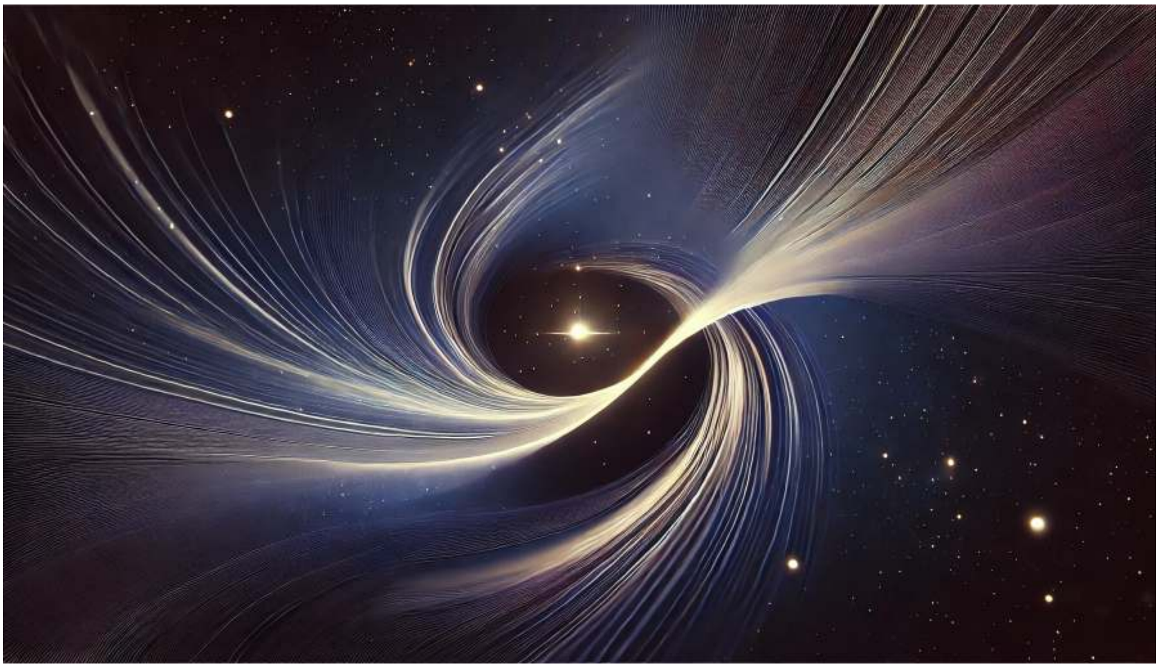

Temporal-Spatial Coupling in Gravitational Lensing: A Reinterpretation of Dark Matter Observations

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Version: v0.7 (Tortola)
First published: 19 December 2025 · Last updated: 10 August 2026
DOI: 10.5281/zenodo.17982540

Abstract

Standard gravitational lensing analysis relies on the Isochrony Axiom—the implicit assumption that the observed image represents a synchronous spatial snapshot of the source. For evolving sources, this approximation breaks down in the presence of conformal metric couplings, creating a "temporal composite" image. This projects temporal depth onto the spatial plane, generating a Temporal-Composite Shear contribution sourced by the underlying Temporal Shear field gradient—arising from gradients in the scalar field's continuous spatial profile (Temporal Topology, TEP)—that is degenerate with gravitational shear in standard static lens reconstructions unless time-domain or variability-dependent observables are included. This contribution forms one component of the Phantom-Mass phenomenology defined here. GW170817 primarily constrains differential propagation and disformal cone tilt; it does not directly test common-mode conformal clock-rate structure along a shared path, although conformal scalar sectors remain indirectly constrained by PPN, equivalence-principle, source-screening, and clock-comparison tests. Because photons and gravitational waves traverse the same path, conformal time dilation is common-mode and cancels in differential measurements. Screening operates via the continuous flattening of Temporal Topology in dense environments, suppressing local field gradients without invoking discrete thin-shell boundaries. Conformal gradients may reproduce specific timing-sensitive aspects of dark-matter-like phenomenology—particularly in the time domain—without violating strong-lens arrival time constraints. Within TEP, the phenomena conventionally attributed to dark matter are real, but the inference of an invisible particulate substance is rejected. TEP interprets the dark sector as Phantom Mass: an apparent convergence, shear, or dynamical mass discrepancy generated when temporal-transport structure is reconstructed under the Isochrony Axiom as synchronous spatial mass. The theory-level claim is stronger than the paper-level evidence claim: within TEP, particulate dark matter is rejected as the fundamental explanation, while this paper tests the lensing-sector realization of that ontology. These results are derived in two regimes: a conservative Reference Envelope (millisecond-scale corrections, directly testable with lensed FRBs) and an Extended Regime (year-scale chronometric corrections and a full lensing-sector dark-sector reinterpretation), whose coherent and source-dependent components are independently testable through blind time-delay residuals, lensing consistency tests, and variability-dependent observables. Within the Extended Regime, where the Isochrony Axiom fails, temporal-field gradients produce an observational degeneracy with particulate dark matter. The Reference Envelope result is the primary, unconditional contribution; the Extended Regime is conditional.

The continuous flattening of the Temporal Topology in dense lensing environments is governed by the abstract environmental operator $\mathcal{S}_\Sigma(\mathcal{E})$. By projecting temporal depth onto the spatial plane, this continuous geometric screening generates the conformal/chronometric component of the Phantom-Mass phenomenology.

Keywords: gravitational lensing – dark matter – modified gravity – cosmology: theory – galaxies: kinematics and dynamics – temporal equivalence principle

1. Introduction

1.1 The Anomaly of the Dark Sector

The existence of dark matter, inferred from gravitational lensing (Walsh et al. 1979), cluster dynamics (Zwicky 1933), and cosmic microwave background observations (Planck Collaboration 2020), represents a significant challenge in modern physics. Despite decades of increasingly sensitive searches, no dark matter particle has been directly detected (Schumann 2019), and tensions persist between cosmological observations at different scales (Riess et al. 2022; Di Valentino et al. 2021). The prevailing paradigm assumes that these anomalies indicate the presence of an invisible *substance*. The question arises whether the apparent mass discrepancy can be resolved by relaxing the *Isochrony Axiom*—the assumption that temporal delays across an image are negligible.

Modified Newtonian Dynamics (Milgrom 1983) and its relativistic extensions (Bekenstein 2004; Skordis & Złośnik 2021) have challenged the dark matter hypothesis by modifying the gravitational force law. However, these approaches typically retain the standard metric assumptions regarding light propagation and causality. The present analysis interrogates a deeper, often unstated assumption underlying the interpretation of all astronomical signals: the nature of simultaneity in image formation.

1.2 The Isochrony Axiom

Gravitational lensing is conventionally treated as the spatial deflection of light rays by mass. Standard practice already models geometric and Shapiro time delays and, for variable sources, the light curves themselves. The closure examined here is narrower and operates one level deeper, in the reconstruction:

The Isochrony Axiom: The closure in which static lens reconstructions treat the source surface-brightness distribution as effectively time-independent across the differential emission-time structure relevant to the reconstruction, after modeled propagation delays have been applied — so that source clocks, photon transport, and observer clocks are mapped onto a single general-relativistic time coordinate.

Stated this way, the axiom is not a claim that lensing analyses ignore time delays. It is the assumption that the residual emission-time structure, once known delays are removed, carries no further information — that the reconstruction may proceed as if the source were synchronous. TEP-GL names this closure and tests its breakdown.

While this axiom serves as a necessary simplification for standard analysis, it breaks down when the differential lookback time across an image becomes comparable to the evolutionary timescale of the source. Standard analysis corrects for the finite speed of light in the arrival time of signals (lookback time) but has typically neglected the differential arrival-time and clock-transfer structure across the image plane. In the presence of generalized metric couplings, this approximation fails. Gravitational lensing is fundamentally an *arrival-time* phenomenon. Images form at the stationary points of the Fermat potential, which encodes the total light-travel time (Blandford & Narayan 1986). In standard General Relativity, the difference in arrival times between images is attributed solely to geometric path differences and the Shapiro delay caused by mass. However, if the coupling between matter and gravity involves a second metric—specifically one that affects matter-clock rates or light-cone tilts—the arrival times of photons can be decoupled from the geometric definitions of "mass" used in standard GR.

If the Isochrony Axiom is violated, an observed "image" is not a snapshot of the source at one moment, but a *temporal composite*. Photons arriving at the same detector time may have left the source at significantly different emission times. For an evolving source, this temporal smearing is mathematically indistinguishable from a spatial distortion (convergence and shear) in a static reconstruction. What is interpreted as "dark matter" may be the projection of this temporal depth onto the spatial image plane.

1.3 The Interpretive Bifurcation

Consider two mathematically equivalent interpretations of the same Fermat potential surface, distinguished only by their treatment of simultaneity:

Interpretation A (Standard Framework): Assumes the Isochrony Axiom. An Einstein ring is analyzed with apparent convergence κ_{obs} exceeding what the visible baryonic mass can produce. A dark matter halo with mass $M_{\text{DM}} = M_{\text{obs}} - M_{\text{baryons}}$ is inferred. The "dark matter" is treated as an unseen substance required to explain the lensing geometry.

Interpretation B (TEP Framework): Rejects the Isochrony Axiom. The observed image is recognized as a *temporal composite*: photons arriving simultaneously at the detector left the source at different emission epochs, with differential delays set by the two-metric structure along each ray. For an evolving source, this temporal depth projects onto the image plane as an apparent spatial distortion. The differential temporal-transfer structure across the lens is computed, and the "excess convergence" is identified as the signature of temporal-field gradients $\nabla(\Delta\tilde{\tau})$ in the lens environment. The "dark matter" is reinterpreted not as a substance, but as the shadow of unmodeled time.

The Critical Point: Both frameworks can reproduce the static lensing observables considered here at the level of the reconstructed Fermat potential. The distinction is not settled by static image geometry alone, but by additional time-domain, variability-dependent, and multi-epoch observables. The difference is not purely observational but *interpretive*—it depends on which axiom (Isochrony vs. TEP) is taken as fundamental.

This bifurcation illustrates that the inference of particulate dark matter from static lensing reconstructions is contingent, in part, on the Isochrony Axiom. If that axiom fails, at least some lensing-inferred dark mass need not correspond to a new substance.

1.4 The Temporal Equivalence Principle (TEP)

The Temporal Equivalence Principle (TEP), introduced in a companion paper (Smawfield 2025a), represents a shift in fundamental perspective, replacing the standard geometric framework with an operational one:

> **Temporal Equivalence Principle (TEP):** The operative physical observables in any non-local measurement are the proper-time intervals registered on physical clock worldlines, together with the phase, frequency, and arrival-time transport connecting those clock events along null signal paths. Under TEP, "gravity" includes the phenomenology of differential clock-transfer structure. The decomposition of this structure into "spatial curvature" (mass) and "temporal dilation" (metric coupling) is gauge-dependent; only the total integrated transport is invariant.

Under TEP, the central question is not "how much mass is bending the light?" but "what is the total emission-to-reception clock-transfer history associated with the signal?" In the conformal-only limit of a two-metric theory, null cones are preserved, meaning the "speed of light" is unchanged, yet the *rate* of proper time accumulation varies. This creates a disconnect between the "gravitational metric" $g_{\mu\nu}$, which carries the gravitational field dynamics and the tensor sector, and the causal "matter metric" $\tilde{g}_{\mu\nu}$, to which all nongravitational matter — test bodies, atomic clocks, and photons alike — couples universally.

The same distinction extends to cosmology. In the canonical TEP interpretation the underlying spatial scale is static, $a_m = 1$, while observed redshift is carried by the conformal clock map $A_{\text{clock}} = (1+z)^{-1}$ (Athens, Paper 26; Thika, Paper 27). Quantities conventionally written as a cosmological scale factor or Hubble evolution may therefore be used as observational or reference variables without implying physical expansion of space. TEP-GL adopts this correspondence: the lensing-sector reconstruction examined here is the local multipath analogue of the cosmological temporal-to-spatial reconstruction treated in those papers.

1.5 Redefining the Dark Sector

Within the Temporal Equivalence Principle, dark matter is not a fundamental particulate substance. The lensing, rotation-curve, and dynamical phenomena conventionally attributed to dark matter are real observational structures, but their interpretation as invisible mass arises from imposing the Isochrony Axiom: the assumption that astronomical images and dynamical reconstructions represent synchronous spatial configurations. TEP replaces this assumption with temporal-transport geometry. Spatially varying clock-transfer structure projects temporal depth into the image plane and dynamical inference pipeline, producing an apparent convergence, shear, and mass discrepancy that standard models absorb as dark matter. Thus TEP does not deny the observations attributed to dark matter; it denies that those observations require a new invisible matter component. This paper tests this claim in the gravitational-lensing sector. It is demonstrated that:

1. **The "dark" signal is a combined temporal-field projection:** Conformal clock-transfer structure contributes chronometric reconstruction, scalar backreaction on $g_{\mu\nu}[\phi]$ and any permitted disformal response generate coherent optical convergence and shear, and source evolution produces the additional Temporal-Composite image response. Standard isochronous reconstruction absorbs these contributions into apparent Phantom Mass.
2. **GW170817 is a differential constraint:** The multi-messenger constraint $|c_\gamma - c_g|/c \lesssim 10^{-15}$ is explicitly reanalyzed. It is shown that this bounds only the *disformal* (cone-tilt) component of the coupling. The *conformal* component, which governs clock rates and drives the phantom-mass phenomenology, is not directly constrained by photon–graviton differential-

propagation bounds because conformal transformations preserve null cones. It remains indirectly constrained by PPN, source-screening, gravitational-redshift, clock-comparison, and equivalence-principle tests.

3. **The Reference Envelope vs. The Extended Regime:** The standard translation of GW170817 timing to propagation-speed bounds is treated as a conservative Reference Envelope for the disformal sector. For the conformal sector, which the GW170817 differential-propagation bound does not directly constrain, the required chronometric amplitudes remain conditionally viable subject to the independent clock, PPN, screening and lensing constraints developed below. The Extended Regime gives the full dark-sector reinterpretation: its chronometric and source-dependent amplitudes are tested by the stated time-domain/variability observables, while its coherent optical component is tested independently through scalar-backreaction/disformal lensing consistency.

By abandoning the Isochrony Axiom, the dark matter problem is reframed as a search for unmodeled temporal structure that has been absorbed into inferred mass. The parameter space where this structure masquerades as dark matter is defined, offering a falsifiable alternative to the particle paradigm.

The TEP thesis holds that the phenomena conventionally attributed to dark matter are the observational projection of temporal-transport geometry, not evidence for a new particulate matter component. The particle interpretation remains a viable effective model under the assumption of synchrony. TEP does not posit an alternative substance; it proposes that the phenomenology traditionally attributed to dark matter is better understood as a metric artifact of the Isochrony Axiom.

This paper does not deny the empirical phenomena conventionally attributed to dark matter. It challenges the inference that those phenomena uniquely require a new particulate matter component. Within TEP-GL, the dark sector is reinterpreted as an observational degeneracy: temporal-field gradients and differential clock-transfer structure can project into lensing reconstructions as apparent convergence and shear when the Isochrony Axiom is assumed. The central question is therefore not whether the observed anomalies exist, but whether they continue to require invisible mass after temporal-composite image formation is included in the forward model.

The claim-discipline framework for the TEP corpus, including the scope limitations of canonical precision tests, is established in TEP-EXP (Paper 9).

2. Theoretical Framework

2.1 The Two-Metric Postulate

The gravitational field dynamics and the matter sector are posited to be governed by distinct metrics related by a scalar field ϕ. The *Gravitational Metric* $g_{\mu\nu}$ carries the gravitational field dynamics and the tensor sector: it has Einstein–Hilbert form and gravitational waves propagate on its null cones. All nongravitational matter — including massive test bodies, atomic clocks, and electromagnetic fields — couples universally to the *causal Matter Metric* $\tilde{g}_{\mu\nu}$, on which nongravitational dynamics, signals, and quantum phases evolve. This universal coupling is the foundational TEP structure (Jakarta, Paper 0); environmental screening can render matter-frame trajectories consistent with the GR limit to high precision. The general two-metric relation (Bekenstein 1993) is adopted:

$$\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu} + B(\phi) \nabla_\mu \phi \nabla_\nu \phi$$

Here, $\tilde{g}_{\mu\nu}$ is the metric on which all nongravitational matter propagates, and $g_{\mu\nu}$ is the metric carrying the gravitational field dynamics. The function $A(\phi)$ defines the *Conformal Sector* (isotropic scaling of proper time/length) and $B(\phi)$ defines the *Disformal Sector* (anisotropic stretching along field gradients). TEP introduces the temporal field as the additional physical degree of freedom: all physical rulers and clocks are built from matter coupled to $\tilde{g}_{\mu\nu}$, while the temporal field's stress-energy and strong-field operators backreact on the Einstein-frame geometry $g_{\mu\nu}$. The structure is therefore not a reinterpretation of measurement alone; sufficiently strong temporal structure induces genuine geometric backreaction (Bahrain, Paper 28), and it is that backreaction — together with the disformal sector — which carries any modification of null-trajectory lensing.

Box 2.0: Sector Dictionary — Three Projections of One Temporal Field

TEP-GL distinguishes three observational projections of the same temporal field. They are complementary and must not be collapsed into a single optical effect. This distinction organizes the remainder of the paper.

1. **Conformal open-path transport** $A(\phi)$: changes the matter-clock transfer associated with geometrically distinct source–observer paths. This can alter arrival-time, distance, velocity, and mass *reconstructions* without changing the unparameterized null trajectories.
2. **Geometric optical response** $g_{\mu\nu}[\phi]$, $B(\phi)$: scalar backreaction on the Einstein-frame metric, together with any active disformal contribution, changes the optical tidal matrix and therefore produces genuine angular convergence and shear.
3. **Temporal-composite response:** for an evolving or moving source, a path-dependent emission-time map produces an additional source-dependent image distortion.

Accordingly, *Phantom Mass* is the apparent non-particulate dark component generated by the combined temporal-field geometry and its reconstruction under the Isochrony Axiom:

$$M_{\text{phantom}}^{\text{TEP}} = \underbrace{M_{\text{rec}}^{A}}_{\text{chronometric reconstruction}} + \underbrace{M_{\text{opt}}^{g[\phi],B}}_{\text{coherent optical}} + \underbrace{M_{\text{TC}}}_{\text{source-dependent}} .$$

These are not three unrelated mechanisms substituted for particulate dark matter. They are three observational projections of one dynamical temporal field. Standard analysis combines them into a single inferred mass distribution precisely because it assumes a universal isochronous mapping between source clocks, photon transport, and observer clocks.

Box 2.1: Relation to Established Scalar-Tensor Theories

TEP is a specific two-metric scalar–tensor effective field theory, distinguished by the universal coupling of all nongravitational matter to a single causal metric and by its dynamical-proper-time interpretation. Its covariant operators draw on established scalar–tensor constructions, while its operational ontology and observable sector decomposition are specific to TEP. The correspondences are:

- **Brans-Dicke Theory:** With $B = 0$, $A(\phi) = e^{\beta_A \phi / M_{\text{Pl}}}$, and the corresponding kinetic and potential choice, the conformal sector contains a Brans–Dicke-like limit in the Jordan frame, with β_A related to the Brans–Dicke parameter ω_{BD}. The correspondence is a limit of the conformal sector, not an identity of the full theory.
- **Horndeski and Beyond:** Higher-derivative scalar-tensor theories (Horndeski, DHOST) provide candidate microscopic completions for the gradient-dependent branch of $\mathcal{S}_\Sigma(\mathcal{E})$. TEP-GL does not commit to a specific completion; the canonical operator is defined at the theory level independently of any particular Lagrangian.
- **Screening and Temporal Topology:** The core TEP framework formulates screening through the environmental operator $\mathcal{S}_\Sigma(\mathcal{E})$ acting on the Temporal Shear Σ_μ, a continuous spatial profile. Imported modified-gravity screening mechanisms are not part of the TEP ontology; the screening object is $\mathcal{S}_\Sigma(\mathcal{E})$ alone.
- **DHOST Theories:** Degenerate Higher-Order Scalar-Tensor theories extend Horndeski by imposing a degeneracy condition that eliminates the additional propagating mode, rather than by keeping the field equations manifestly second-order. TEP is compatible with this broader class but does not require it.

Key Distinction: TEP's distinguishing content is the universal causal-metric coupling together with the *observational interpretation* it licenses — specifically, the recognition that conformal coupling creates a "temporal composite" image whose residual timing structure a standard reconstruction absorbs as mass. Individual covariant operators are drawn from established scalar–tensor constructions; the sector dictionary, the universal coupling, and the resulting lensing phenomenology are specific to TEP.

2.2 The Chronometric Lensing Framework

TEP-GL lensing contains three projections of one temporal field: coherent optical response through scalar backreaction and any permitted disformal contribution; conformal open-path chronometric reconstruction; and the source-dependent Temporal-Composite response.

1. The Static Clock-Transfer Contribution (Chronometric Reconstruction)

The conformal clock sector modifies the relation between accumulated matter-frame time and the geometry reconstructed under an isochronous model. The associated effective clock-transfer contrast is:

$$\Delta\tilde{\tau}_{\text{static}} = \frac{1}{c}\int (A(\phi) - 1)\, dl$$

In the phenomenological isochronous reconstruction used here, this is represented as a source-independent correction to the inferred arrival-time surface. In multipath configurations, it is therefore operationally degenerate with the mass normalization inferred from multipath timing, distance, and joint image–delay reconstruction. The angular positions of Einstein rings and major arcs remain governed by the optical geometry.

The expression above is an effective matter-clock transfer functional evaluated in the chosen clock congruence; it is not an additional photon proper-time or geometric null-travel-time delay. In the pure-conformal limit, its operational content must be defined through emission/reception clock calibration and frequency transport. Genuine new path-dependent null delay beyond the GR geometry arises only through gravitational backreaction, the disformal sector, or another explicitly derived non-exact contribution. As established in Axiom 1, the conformal limit preserves null cones: pure multiplication by $A^2(\phi)$ does not bend a photon onto a new null trajectory. Any genuinely new static null-trajectory bending beyond the conformally related geometry must arise through gravitational backreaction in $g_{\mu\nu}$, a non-negligible disformal contribution from $B(\phi)$, or another explicitly derived part of the coupled field solution. The observable lensing signal in the conformal sector is therefore a clock-transfer discrepancy — a difference between the matter-frame time accumulated along each path and the time inferred from an isochronous GR lens model — rather than a direct refraction of null geodesics.

2. The Dynamic Shutter (Temporal Lensing)

While the static term shifts the arrival-time surface, the gradient of the differential delay field $\nabla(\Delta\tilde{\tau})$ acts as a "shutter" that modulates the arrival time of photons from different parts of the source. For a source with evolution or motion, this creates a *Temporal-Composite Shear*:

$$\gamma^{\text{TC}} \propto \left(\frac{\partial I}{\partial t} + \vec{v}_s \cdot \nabla I\right)\nabla(\Delta\tilde{\tau})$$

The nomenclature follows the canonical corpus convention: *Temporal Shear* denotes the field gradient $\Sigma_\mu \equiv \nabla_\mu \ln A(\phi)$, which generates the path-dependent clock-transfer field, whereas *Temporal-Composite Shear* γ^{TC} denotes the observable source-dependent image response to that field. The two must not be conflated: the former is a property of the temporal field, the latter of the observation.

This *Dynamic Shutter* explains the anomalies—the "phantom mass" that appears to fluctuate with source type. It operates as a re-ordering of wavefronts in time rather than a deflection of rays. Crucially, this effect applies even to "static" sources (like elliptical galaxies) due to their proper motion $\vec{v}_s$ across the delay gradient. (See Section 3 for the full derivation).

Together, these two mechanisms constitute the TEP framework: the clock-transfer contribution may contribute to the phenomenology conventionally attributed to the dark matter halo, and the dynamic shutter may account for part of the apparent "complexity" of substructure. Any genuinely new null-trajectory bending beyond the conformally related geometry must arise through gravitational backreaction in $g_{\mu\nu}$, a non-negligible disformal contribution, or another explicitly derived part of the coupled field solution.

2.3 Operational Axioms: The TEP Framework

The TEP framework rests on four foundational axioms. These are not approximations or perturbative corrections to General Relativity; they constitute a complete operational framework for interpreting gravitational phenomenology. They are adopted as *primary postulates* from which observational consequences are derived:

Axiom 1 (Causal Universality): In the Conformal Limit ($B = 0$), the null cones of $g_{\mu\nu}$ and $\tilde{g}_{\mu\nu}$ are identical. Photons and gravitational waves follow the same null geodesics. No "speed of light" difference exists in this limit. *This axiom is exact, not approximate.*

Axiom 2 (Proper Time Primacy): The fundamental observables in any timing measurement are the proper-time intervals $\Delta\tilde{\tau}$ registered on physical emitter and receiver clock worldlines, together with the phase, frequency, and arrival-time transport connecting those clock events along $\tilde{g}$-null signal paths. Coordinate time differences Δt are not observables; they are inferred quantities dependent on the metric model. Null proper time along the photon trajectory is identically zero (Jakarta); wherever this paper refers to accumulated path time, the phrase denotes the operational clock-transfer functional associated with the null path, not proper time experienced by the photon. *This axiom establishes clock-registered proper time as the irreducible physical observable; all other timing quantities are derived.*

Axiom 3 (Non-local temporal transport): TEP admits two distinct classes of non-local timing observable.

(a) Synchronization holonomy: A genuinely closed, direction-reversing transport loop can exhibit residual non-closure only when the transport connection contains non-exact structure, such as the disformal $B(\phi)$ sector. In the pure conformal limit, the $A(\phi)$ contribution is exact and its residual closed-loop integral vanishes.

(b) Differential path transport: Distinct open propagation paths may accumulate different temporal-transfer corrections. In gravitational lensing, two images correspond to two open paths γ_i, γ_j, and the observable is the differential transport residual

$$\Delta\mathcal{T}_{ij}^{\text{resid}} = [\mathcal{T}[\gamma_i] - \mathcal{T}[\gamma_j]] - [\mathcal{T}_{\text{GR}}[\gamma_i] - \mathcal{T}_{\text{GR}}[\gamma_j]],$$

where $\mathcal{T}[\gamma] \equiv \Delta\tilde{\tau}[\gamma]$ is the time-transport functional. This is an *open-path blind-prediction residual*, not a closed-loop synchronization holonomy. Because each image has a single observed arrival time, algebraic closure of measured pairwise delays around image triplets vanishes identically. This does not cause the algebraic sum of observed image delays to fail to close; it alters the relation between lens geometry and observed arrival time without violating the algebraic identity among the observed arrival times.

This axiom distinguishes open-path differential residuals (the GL observable) from closed-loop holonomy (the domain of triangle time-transfer and direction-reversing experiments), and is consistent with the refined strong-lensing formulation in TEP-LENS (Paper 19).

Measurement Protocol for the GL Observable. The lensing-sector observable is not algebraic delay closure. It is a blind-prediction residual. For each image pair (i, j), compare the observed delay $\Delta t_{ij}^{\text{obs}}$ with the pre-specified GR lens-model prediction $\Delta t_{ij}^{\text{GR}}$:

$$R_{ij} = \Delta t_{ij}^{\text{obs}} - \Delta t_{ij}^{\text{GR}}.$$

TEP predicts that these residuals should correlate with temporal-transport tracers such as projected potential depth, magnification, variability timescale, or other lens-environment proxies. Algebraic closure of observed pairwise delays is not a TEP discriminator, because it vanishes identically whenever each image has a unique arrival time.

Systematic Error Budget: Several astrophysical systematics can produce apparent residuals that must be distinguished from true temporal-transport effects:

- **Lens Model Degeneracies:** The mass-sheet degeneracy and source-position transformation can bias predicted delays. These affect the *predicted* delay (from the model), not the *observed* delay. The test compares observed delays against model predictions, and systematic lens-model error must be propagated into the residual uncertainty.
- **External Convergence:** Line-of-sight structure acts approximately as an external mass-sheet transformation, rescaling the Fermat-potential and time-delay normalization. If omitted, it can generate coherent biases in R_{ij}. Each lens must therefore marginalize over κ_{ext} and the associated mass-sheet uncertainty before testing for a TEP residual.
- **Microlensing:** Stellar microlensing in the lens galaxy can shift apparent arrival times by hours to days. However, microlensing is stochastic and uncorrelated between images; over an ensemble of lens systems, microlensing-induced residuals should average to zero with RMS scaling as $1/\sqrt{N}$.
- **Host Galaxy Delays:** Differential extinction or scattering in the host can introduce chromatic delays, but TEP's temporal-transport residual is achromatic. Chromatic residual patterns indicate astrophysical contamination, not metric effects.

Discriminator: True temporal-transport residuals should produce a *systematic* non-zero regression coefficient or covariance between R_{ij} and a pre-specified temporal-transport predictor, using a fixed image-pair ordering and hierarchical marginalization over lens-model degeneracy, κ_{ext}, microlensing, and host-galaxy delays, while astrophysical systematics produce *random* residuals that average to zero across multiple systems. An unconditional mean could cancel even when a real path-dependent correlation exists. The correlation is the correct TEP discriminator.

Axiom 4 (Screening and Temporal Topology): Screening manifests as a continuous spatial profile (Temporal Topology) governed by the non-linear superposition of field gradients (Temporal Shear), suppressing fifth forces and lensing anomalies in dense environments while leaving cosmology accessible to dynamics. The suppression of local Temporal Shear in deep potential environments continuously reduces

$$\Sigma_\mu^{\rm obs} = \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu.$$

Channel-specific PPN responses are evaluated only after this environmental projection is applied. Existing multi-messenger constraints (e.g., GW170817) are interpreted within the standard framework that assumes a single metric governs all sectors; in the TEP framework, these constraints apply to specific parameter combinations (primarily the disformal sector) and do not constitute blanket exclusions of two-metric effects. *This axiom establishes that environmental suppression is a geometric property of the field's spatial profile, not a binary boundary condition, and that constraints derived under Isochrony-assuming frameworks must be re-derived operationally within TEP before being applied as exclusions.*

These axioms are mutually consistent and together define the TEP interpretation of gravitational phenomenology. They replace the implicit Isochrony Axiom of standard lensing analysis with an explicit dynamical-time framework.

The Hidden Closure in Standard Practice: In addition to the Isochrony Axiom, most observational inference quietly assumes that time transport is globally integrable: after correcting for known effects (Sagnac, Shapiro, troposphere, etc.), a single global time coordinate can be assigned such that closed-loop synchronization holonomy vanishes. Operationally, this is the step that licenses the non-local conversion $d = c\,t$ and the inference “timing residual $\Rightarrow$ mass residual.” In TEP, this closure is not assumed; Axiom 3 treats open-path differential residuals as the primary observables, while true synchronization holonomy requires direction-reversing closed loops or non-exact transport structure.

2.4 Conformal vs. Disformal Phenomenology

The distinction between the two sectors is critical for interpreting multi-messenger constraints, and it rests on a fundamental distinction between *Single-Path* and *Multipath* measurements. Furthermore, it necessitates a redefinition of the "speed of light":

- **Conformal Sector ($A(\phi)$):**
  - **Geometry:** Preserves angles and null cones. $\tilde{g}_{\mu\nu}k^\mu k^\nu = A^2(\phi)g_{\mu\nu}k^\mu k^\nu = 0$.
  - **Local Invariance vs. Global Variability:** Local c remains invariant (measured as $299,792,458$ m/s by any local clock). What varies globally is not the local speed of light, but the *inferred ratio between spatial separation and matter-clock transfer registered between endpoints* along extended paths. Because the rate of proper time accumulation $d\tilde{\tau} = A(\phi)d\tau_g$ varies with location, the time required to traverse a fixed spatial interval depends on the scalar field value. To an observer assuming a universal clock, light appears to speed up or slow down depending on the path.
  - **Single-Path Physics (GW170817):** Photons and gravitational waves from the exact same source coordinate follow the *same* null geodesic. Any temporal distortion $A(\phi)$ along this path is common-mode. The signals do not diverge because they share the same history.
  - **Multipath Physics (Lensing):** Gravitational lensing involves light rays taking *different* paths around a mass distribution. These paths traverse different regions of the scalar field $\phi(\vec{x})$. The differential clock-transfer structure between these paths generates the chronometric component of the Phantom-Mass signature; coherent optical convergence and shear arise through scalar backreaction and any permitted disformal response.
- **Disformal Sector ($B(\phi)$):**
  - **Geometry:** Tilts null cones. The effective speed of light differs from the speed of gravity: $c_\gamma \neq c_g$.
  - **Observables:** Differential arrival times between species, constrained by GW170817 to $|c_\gamma - c_g|/c \lesssim 10^{-15}$ for the path-averaged monopole.

2.5 The "Phantom Mass" Mechanism

Standard lensing reconstruction solves for a mass distribution $\Sigma(\vec{\theta})$ that reproduces the observed image distortions. This reconstruction assumes the Isochrony Axiom: $I_{\rm obs}(\vec{\theta}) = I_{\rm src}(\vec{\beta})$. However, in the TEP framework, the observed image at a given observation time $t_{\rm obs}$ is the source evaluated at a shifted effective time:

$$I_{\rm obs}(\vec{\theta},\, t_{\rm obs}) = I_{\rm src}\Big[\vec{\beta}_{\rm opt}(\vec{\theta}),\; t_{\rm obs} - \Delta T_{\rm eff}(\vec{\theta})\Big],$$

where $\Delta T_{\rm eff}(\vec{\theta})$ is the effective clock-transfer contrast along the line of sight at image position $\vec{\theta}$, and $\vec{\beta}_{\rm opt}(\vec{\theta})$ is the optical ray map generated by the scalar-backreacted and disformal geometry. For a finite exposure over $[t_1, t_2]$, the recorded image is the time-averaged quantity:

$$\bar{I}_{\text{obs}}(\vec{\theta}) = \frac{1}{\Delta t_{\text{exp}}} \int_{t_1}^{t_2} I_{\text{src}}\left[\vec{\beta}_{\text{opt}}(\vec{\theta}),\ t - \Delta T_{\text{eff}}(\vec{\theta})\right] dt.$$

For a spatially varying conformal coupling, ΔT_{eff} varies across the image plane. If the source has temporal variability (secular evolution, rotation, or fluctuations) on the timescale of $\nabla_\theta(\Delta T_{\text{eff}})$, the finite-exposure average smears the recorded image.

The Equivalence: A gradient in arrival time across an image is mathematically equivalent to a shearing of the source frame. To a static observer assuming isochrony, this "temporal shear" is indistinguishable from the "gravitational shear" caused by mass. Thus, *purely temporal structure is misinterpreted as "Phantom Mass" (dark matter).*

2.6 Two Regimes of TEP-GL

Box 1: The Two Observational Regimes

The TEP-GL framework operates in two distinct regimes, distinguished by the magnitude of the differential clock-transfer residual $\Delta\tilde{\tau}$ across the lens:

Regime I: The Reference Envelope

- **Assumption (Standard):** The GW170817 multi-messenger timing constraint applies to all metric sectors equally.
- **Constraint basis:** The standard translation of timing to propagation-speed bounds ($\lesssim 10^{-15}$).
- **Delay scale:** $\Delta\tilde{\tau} \sim 10^{-3}$--$1$ s (milliseconds to seconds) on halo scales.
- **Primary observables:** Time-domain signatures in rapidly varying sources—lensed FRBs, GRBs.
- **Dark matter status:** TEP is a *precision systematic* in time-delay cosmography, not a wholesale DM replacement.
- **Falsification:** Null detection of achromatic timing residuals at < 0.1 ms excludes this regime.

Regime II: The Extended Regime

- **Assumption (Sector Decoupling):** GW170817 constrains only differential disformal coupling; common-mode conformal temporal structure is not directly constrained by it, though it remains bounded by clock, PPN, and redshift channels (see Axiom 4).
- **Delay scale:** $\Delta\tilde{\tau} \sim 1$--$10$ years, driven by the conformal factor $A(\phi)$ integrated over halo scales (Mpc).
- **Primary observables:** The full phenomenology of "dark matter" in lensing — cluster arcs, cosmic shear — arising from coherent scalar backreaction on $g_{\mu\nu}[\phi]$ and any permitted disformal response, combined with the conformal clock-transfer reconstruction. The Dynamic Shutter supplies the additional source-dependent component.
- **Dark matter status:** Phantom Mass is the apparent spatial component generated when temporal-field geometry, chronometric transport, and Temporal-Composite response are reconstructed under the Isochrony Axiom. The underlying physical entity is the temporal field ϕ, not particulate dark matter.
- **Falsification:** Null source-variability correlations exclude the Temporal-Composite channel; CMB–galaxy consistency constrains its amplitude, while the coherent scalar-metric sector is tested independently.

This work demonstrates that *the Extended Regime (Regime II) is a viable physical alternative*. The Reference Envelope (Regime I) is a useful conservative baseline for calibration, but it represents a scenario where the temporal field is suppressed to match constraints whose differential-propagation interpretation does not directly bound the common-mode conformal clock sector.

2.7 Why Lensing May Not Be Purely Spatial

Standard gravitational lensing analysis contains a simplifying temporal assumption that has typically been treated as exact. This assumption is made explicit, and its breakdown under TEP leads directly to the dark matter reinterpretation.

The Hidden Assumption in Image Formation

When a lensed galaxy is observed, photons are collected over an exposure time and an "image" is reconstructed. This reconstruction implicitly assumes:

1. All photons arriving during the exposure left the source at approximately the same epoch
2. Differences in arrival angles correspond to differences in spatial paths, not temporal paths
3. The reconstructed "shape" represents a spatial snapshot of the source at one moment

These assumptions constitute the *Isochrony Axiom* applied to image formation. Standard analysis corrects for the mean lookback time (distance), but assumes that the variance in lookback time across the image is negligible. This overlooked variance is defined as *Differential Lookback Time*. In a two-metric framework, this variance is an additional source-dependent reconstruction residual.

The Temporal Composite Mechanism

Consider an extended galaxy being lensed. Light from different parts of the galaxy:

- Leaves the source at different times (the source is evolving on Myr timescales)
- Takes different paths through the lens (different impact parameters)
- Accumulates time differently through regions with different $A(\phi)$ values
- Arrives at the detector at the "same" observation time

If $A(\phi)$ varies spatially—forming a halo-like configuration around the lens—then:

$$\Delta t_{\text{path}} = \int_{\text{path}} \frac{A(\phi)}{c}\, dl$$

differs between rays at different impact parameters. For halo-scale propagation distances $L \sim 2$ Mpc and conformal variations $\Delta A/A \sim 10^{-6}$, the differential delay is:

$$\Delta t \sim \frac{\Delta A}{A} \cdot \frac{L}{c} \sim \text{years}$$

This delay is **consistent with observed strong lensing time delays**. However, it implies that the "Dynamic Shutter" effect (temporal smearing) is negligible for slowly evolving galaxies. For slowly evolving galaxies the source-dependent Dynamic Shutter is negligible. The dominant signal is therefore the coherent static temporal-field sector: physical optical shear/convergence arises from scalar backreaction and any permitted disformal response, while the conformal sector supplies the associated chronometric reconstruction. The temporal smearing (Mechanism B) becomes the dominant signal only for fast transients (FRBs).

Why This Creates "Phantom Mass"

For a source that evolves on timescales comparable to Δt:

- If the source was more compact in the past → inner regions (earlier epoch) appear smaller
- If the source was less compact in the past → inner regions appear larger
- The reconstructed "shape" is systematically distorted by temporal mixing

Standard lensing analysis interprets *any* systematic shape distortion as evidence for mass (convergence and shear). The temporal smearing is mathematically indistinguishable from gravitational lensing distortion. What has been interpreted as "dark matter" may be, in whole or in part, *temporal depth projected onto the spatial image plane*.

Why This Effect Is Achromatic

The conformal factor $A(\phi)$ rescales proper time identically for all photon frequencies. Unlike plasma dispersion (which scales as ν^{-2}) or dust extinction (wavelength-dependent), conformal temporal coupling is *perfectly achromatic*. General relativistic lensing is likewise achromatic, so achromaticity alone does not discriminate a temporal-field contribution from ordinary gravitational lensing. The operational significance is the converse: because the TEP contribution is achromatic, it cannot be separated from GR lensing by colour information, but it *can* be separated from plasma and dust systematics — which is precisely what makes multi-frequency timing the discriminating measurement (Section 5.1).

The Central Thesis

Under TEP, the existence of "dark matter" is inferred rather than directly observed. It is a *conclusion contingent on the Isochrony Axiom*. If that axiom fails—if the universe is not temporally synchronous in the way standard analysis assumes—then the inferred "dark mass" can be modeled as unmodeled temporal structure. The dark sector is modeled not as a substance, but as the shadow of time. No claim is made regarding uniqueness of the two-metric realization, only the operational equivalence of temporal gradients to inferred mass under isochrony.

3. The Phantom Mass Mechanism

3.1 The Canonical Disformal Metric and Null-Trajectory Structure

The Phantom Mass mechanism is derived from the canonical TEP matter metric. The foundational two-metric structure (Jakarta, Paper 0) relates the gravitational metric $g_{\mu\nu}$ to the causal matter metric $\tilde{g}_{\mu\nu}$ through the disformal map:

$$\tilde{g}_{\mu\nu} = A^2(\phi)\, g_{\mu\nu} + B(\phi)\, \nabla_\mu\phi\, \nabla_\nu\phi$$

where $A(\phi)$ is the universal conformal coupling and $B(\phi)$ is the disformal coupling. This is the canonical metric $A^2 g_{\mu\nu} + B\, \nabla_\mu\phi\, \nabla_\nu\phi$, not a Newtonian-gauge metric with a conformal factor inserted only in the spatial sector.

3.1.1 Conformal Invariance of Null Trajectories

In the purely conformal subclass ($B = 0$), the matter metric reduces to $\tilde{g}_{\mu\nu} = A^2(\phi)\, g_{\mu\nu}$. A standard result of conformal geometry in four dimensions is that null geodesics are preserved: if k^μ is tangent to a null geodesic of $g_{\mu\nu}$, the same path is a null geodesic of $\tilde{g}_{\mu\nu}$ with a rescaled affine parameter. Therefore:

$$B = 0 \quad \Longrightarrow \quad d\tilde{s}^2 = 0 \quad \Longleftrightarrow \quad ds^2 = 0.$$

This has three immediate consequences for gravitational lensing:

1. A^2 alone does not change the null trajectory. Pure conformal rescaling preserves null cones; it cannot generate a spatial refractive index or bend photons onto new paths.
2. Static geometric bending beyond the conformally related geometry must come from $g_{\mu\nu}[\phi]$ backreaction — the scalar field modifying the Einstein-frame metric through the coupled field equations — or from $B \neq 0$, which tilts the matter null cone relative to the gravitational null cone.
3. The conformal sector can alter endpoint clock calibration, frequency transport, and timelike source evolution. It acts directly on timelike observables (orbital periods, spectroscopic velocities, standard-candle distance calibration) but cannot generate the printed spatial refractive index $n_{\rm eff} \simeq 1 - 2\Psi + \alpha(\phi)$ that would follow from inserting $A^2(\phi)$ into the spatial sector of a Newtonian-gauge metric alone.

3.1.2 Separating Null and Timelike Observables

The lensing deflection angle decomposes into two physically distinct channels:

$$\delta\theta_{\rm lens} = \delta\theta_{g[\phi]} + \delta\theta_B$$

where $\delta\theta_{g[\phi]}$ is the deflection from scalar backreaction on the geometric metric $g_{\mu\nu}$ — the scalar field modifying the Einstein-frame curvature through the coupled Einstein–scalar equations — and $\delta\theta_B$ is the deflection from the disformal term $B(\phi)\nabla_\mu\phi\nabla_\nu\phi$, which tilts the matter null cone along the field gradient. The conformal factor $A(\phi)$ does not appear in $\delta\theta_{\rm lens}$ at leading order because it cancels from the null condition.

The worked backreaction example is the scalar-Gauss-Bonnet (sGB) coupling studied in Bahrain (Paper 28, Appendix L). In shift-symmetric sGB, the scalar field $\phi \sim Q_s/r$ backreacts on the geometric metric at $\mathcal{O}(\eta^2)$ (where $\eta = 3\alpha_{\rm GB}/M^2$), producing a genuine correction to the null-trajectory bending. The conformal factor $A = e^{-\phi}$ modifies clock rates at $\mathcal{O}(\eta)$ — acting on timelike observables such as the ISCO — but does not contribute to the shadow at leading order because null geodesics are conformally invariant. Bahrain Appendix L explicitly isolates this sGB backreaction contribution and demonstrates the different coupling orders of the null and timelike observables: the shadow is sensitive to the geometric metric at $\mathcal{O}(\eta^2)$ while the ISCO feels the conformal factor at $\mathcal{O}(\eta)$.

3.1.3 The Temporal Composite Observable

The conformal sector produces observable effects through timelike channels: endpoint clock calibration, frequency transport, and source evolution. These are captured by the temporal composite observable, which relates the observed image intensity to the emission-time shift induced by the temporal field:

$$\delta I_{\rm obs} \simeq -\frac{\partial I_{\rm src}}{\partial t_{\rm em}}\, \delta t_{\rm em}$$

where $\delta t_{\rm em}$ is the differential emission-time shift between paths, sourced by the complete metric $\tilde{g}_{\mu\nu}$ rather than an unmatched spatial conformal factor. The delay must come from the full disformal metric — including $g_{\mu\nu}[\phi]$ backreaction, the disformal term B, endpoint clock-rate differences through $A(\phi)$, and frequency transfer — not from a path integral $\int A(\phi)\,dl/c$ interpreted as photon proper-time accumulation. Null proper time is zero (Jakarta); any measured open-path effect must be formulated through emission and reception clocks, actual coordinate delay from the geometric and disformal metric, or frequency transfer.

The static clock-transfer contribution from the conformal sector modifies the relation between accumulated matter-frame time and the geometry reconstructed under an isochronous model. The associated effective clock-transfer contrast along a path γ is:

$$\Delta\tilde{\tau}_{\rm static} = \frac{1}{c}\int_\gamma \big(A(\phi) - 1\big)\,dl$$

This is a clock-transfer discrepancy — a difference between the matter-frame time accumulated along each path and the time inferred from an isochronous GR lens model — not a direct refraction of null geodesics. It contributes a source-independent term to the arrival-time (Fermat) surface and is operationally degenerate with the mass normalization inferred from multipath timing, distance, and joint image–delay reconstruction. The angular positions of Einstein rings and major arcs are governed by the optical geometry; any TEP modification of those positions arises through scalar backreaction on $g_{\mu\nu}$, the bounded disformal sector, or a source-dependent temporal-composite displacement. The path integral here represents accumulated clock-rate difference along the open path, not photon proper time.

3.1.4 The Amplification Matrix and Jacobian Decomposition

Integrating the geodesic deviation equation along the line of sight yields the amplification matrix $\mathcal{A}_{ij}$, which maps source-plane displacements to image-plane displacements:

$$\mathcal{A}_{ij} = \frac{\partial\beta_i}{\partial\theta_j} = \delta_{ij} - \int_0^{\chi_s} \frac{(\chi_s - \chi)\chi}{\chi_s}\,\mathcal{R}_{ij}(\chi)\,d\chi$$

where $\mathcal{R}_{ij}$ is the optical tidal matrix constructed from the Riemann tensor of the geometric metric $g_{\mu\nu}[\phi]$, projected onto the screen space. The TEP correction to the optical tidal matrix comes from scalar backreaction on $g_{\mu\nu}$ and from the disformal sector, not from a spatial conformal factor:

$$\mathcal{R}_{ij} = \mathcal{R}_{ij}^{(\Psi)} + \mathcal{R}_{ij}^{(g[\phi])} + \mathcal{R}_{ij}^{(B)}$$

where $\mathcal{R}_{ij}^{(\Psi)}$ is the standard GR contribution, $\mathcal{R}_{ij}^{(g[\phi])}$ is the scalar-backreaction correction to the geometric metric, and $\mathcal{R}_{ij}^{(B)}$ is the disformal correction. The conformal factor does not appear at leading order because it cancels from the null trajectory.

The magnification of an image is $\mu = (\det\mathcal{A})^{-1}$. To first order in the TEP correction $\delta\mathcal{A}_{ij}$, the fractional change in magnification is:

$$\frac{\delta\mu}{\mu} = \mathrm{Tr}\left(\mathcal{A}_{\rm GR}^{-1}\,\delta\mathcal{A}\right)$$

This defines the lensing amplification kernel $\mathcal{P}_\mu$, a projection operator that maps the scalar-backreaction and disformal Hessian to the observable magnification shift. Near a critical curve, where $\det\mathcal{A}_{\rm GR} \to 0$, the kernel diverges as $\mu_{\rm GR}^2$, amplifying small corrections into large observable residuals. Because the inverse reconstruction is correspondingly sensitive to small timing and geometric perturbations there, this supplies a candidate mechanism capable of closing the factor-of-$\sim$90–750 amplitude gap between direct potential-sampling and observed delay shifts identified in Paper 19 (SN Refsdal) — locating that discrepancy in Jacobian amplification rather than in a free phenomenological coefficient. A numerical resolution requires evaluation of the complete tensor and lens-model response, not the scalar log-magnification proxy alone; that calculation is not performed here.

Formally, the operational response proxy used in the blind-prediction tests (Paper 19, §2.3) is the first-order truncation of this kernel:

$$\Gamma_t(i) = 1 + \kappa_{\rm lens}\,\mathcal{P}_\mu\big[\nabla\phi\big](i) \;\approx\; 1 + \kappa_{\rm lens}\log_{10}\!\big(\mu_{\rm norm}(i)\big)$$

This is a phenomenological scalar truncation of the tensor response. Weak shear alone does not make it exact; exact reduction additionally requires an approximately isotropic response, $\delta\mathcal{A}_{ij} \propto \delta_{ij}$. The log-magnification form follows from $\delta\mu/\mu = \mathrm{Tr}(\mathcal{A}^{-1}\delta\mathcal{A})$ when $\delta\mathcal{A}$ is proportional to the identity. The mu–kappa–gamma systematic is the residual error from truncating the full tensor kernel to a scalar log-magnification proxy. Eliminating it requires direct evaluation of $\mathcal{P}_\mu$ from high-resolution mass models, which is the definitive next phase identified in Paper 19.

Response-coefficient convention. Following the corpus rule established in TEP-COS (Paper 10) for $\kappa_{\rm MSP}$, the quantity $\kappa_{\rm lens}$ is an *observable channel response coefficient*: it measures how strongly the lensing reconstruction responds to Temporal Shear. It is not the bare microscopic conformal coupling β_A, not $A-1$, and not the locally active PPN coupling, and it must not be numerically identified with the amplitudes of other channels — such as $\epsilon_T^{\rm HC}$ (TEP-HC) or the line-of-sight amplitude of TEP-C0 — unless a solved environmental transfer function establishes that identification. Schematically, the lensing-channel observable takes the form $\Delta O_{\rm GL} = \kappa_{\rm GL}\,\mathcal{S}_{\rm GL}(\mathcal{E})\,\mathcal{F}_{\rm GL}[\Delta\ln A, \Sigma_\mu, C_A;\ \Phi, \rho, z]$, with the environmental screening operator applied before comparison with any other channel.

Amplitude dictionary. The manuscript refers to several distinct quantities that must not be conflated:

- β_A: the microscopic conformal parameter (bare Lagrangian quantity).
- $\Sigma_\mu = \nabla_\mu \ln A(\phi)$: the underlying Temporal Shear.
- $\Sigma_\mu^{\rm obs} = \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu$: the observable environmentally projected Temporal Shear.
- $\langle\Delta A\rangle_\gamma = \int_\gamma \Sigma_\mu^{\rm obs}\,dx^\mu$: the open-path clock-transfer contrast.
- $\kappa_{\rm lens}$: the lensing reconstruction response coefficient relating temporal-field structure to inferred convergence.

These quantities are not numerically interchangeable without a solved environmental transfer function.

3.1.5 Interpretation: Geometric vs. Temporal Contributions

The Jacobian decomposition reveals two physically distinct contributions to image distortion:

Term	Source	Physical Origin	Observational Signature
Geometric	$\Psi_{,ij}$	Spatial curvature from mass	Standard convergence κ and shear γ
Scalar backreaction	$g_{\mu\nu}[\phi]$	Scalar field modifying Einstein-frame curvature	Coherent tangential shear mimicking DM halo
Disformal	$B(\phi)\nabla_\mu\phi\nabla_\nu\phi$	Null-cone tilt along field gradient	Direction-dependent deflection
Temporal Composite	$\mu_s^i\,\partial_j\Delta T_{\rm eff}$	Source motion × clock-transfer delay gradient	Stochastic shear noise correlated with kinematics

The scalar-backreaction and disformal terms produce coherent contributions to the shear field. The temporal composite term arises when the source position evolves during the differential clock-transfer delay across the image; for a source with proper motion $\vec{\mu}_s$, the effective source position becomes:

$$\beta^i_{\rm eff}(\vec\theta) = \beta^i_{\rm geom}(\vec\theta) - \mu_s^i\,\Delta T_{\rm eff}(\vec\theta)$$

where μ_s^i is the i-th component of the source proper motion and $\Delta T_{\rm eff}$ is the scalar clock-transfer contrast. The corresponding Temporal-Composite contribution to the Jacobian is:

$$\delta\mathcal{A}^{\rm TC}_{ij} = -\mu_s^i\,\partial_j\Delta T_{\rm eff}$$

This adds an asymmetric, source-dependent contribution to the Jacobian that does not average coherently but increases the variance of shear measurements.

Summary: The Phantom Mass Decomposition

The full amplification matrix in TEP is:

$$\mathcal{A}_{ij} = \underbrace{\mathcal{A}_{ij}^{\mathrm{GR}}}_{\text{Baryonic Lensing}} + \underbrace{\mathcal{A}_{ij}^{(g[\phi],\mathrm{static})}}_{\text{"Dark Matter" (Coherent)}} + \underbrace{\mathcal{A}_{ij}^{(\mathrm{dyn})}}_{\text{Temporal Composite (Stochastic)}}$$

Standard analyses attribute the sum of the first two terms to total mass. TEP identifies the second term as the coherent optical component of Phantom Mass — a geometric effect of scalar backreaction on the Einstein-frame metric and disformal null-cone tilt, not particulate matter. The third term provides the unique observational discriminator: excess shear dispersion correlated with source kinematics. The conformal factor $A(\phi)$ acts on timelike observables and clock calibration but does not generate the null-trajectory bending at leading order.

Box 3.1: A Minimal Toy Model Estimate (Halo Scale Integration)

To demonstrate the order of magnitude, consider a simple spherical conformal halo profile:

$$A(\phi) = 1 + \epsilon \ln(r/r_0)$$

For a coupling strength $\epsilon \approx 10^{-6}$ and a characteristic scale $r_0 = 10$ kpc:

- **Integration Path:** The delay is integrated only over the effective halo depth ($L_{halo} \approx 2$ Mpc), not the full cosmological path. This respects the locality of the potential well.
- **Differential Clock-Transfer Reconstruction Scale:** Across an Einstein radius ($r_E \approx 5$ kpc), the differential clock-transfer contrast corresponds to:

$$\Delta\tilde{\tau} \sim \frac{\epsilon}{2}\frac{L_{halo}}{c} \approx \frac{10^{-6}}{2} \cdot (6.5 \times 10^6 \text{ light-years}) \approx 3.2 \text{ years}$$

- **Consistency:** This ~3 year scale is commensurate with observed time delays in strong lens systems (e.g., SN Refsdal), removing the earlier millennia-scale inconsistency and placing the Extended-Regime clock-transfer scale in the observed strong-lens range. It is an effective matter-clock reconstruction scale, not additional photon proper time.
- **Mechanism A (Clock-Transfer Reconstruction):** The static gradient $\nabla(\Delta\tilde{\tau})$ produces unmodeled differential clock-transfer along existing lens paths. When a GR lens model is required to reproduce a path-dependent timing map while assuming isochrony, it can absorb this residual into inferred convergence, mass-sheet normalization, or distance calibration — the reconstruction-space "Phantom Mass" signature. It does not refract null geodesics onto new trajectories; the conformal sector preserves null cones. The corresponding physical optical-tidal contribution is carried by the scalar-backreaction and disformal channels of §3.1.4.
- **Mechanism B (Stochastic):** The dynamic shutter effect $\vec{\mu}_s \cdot \nabla\tau$ is small for galaxies on year-timescales, but dominant for millisecond transients (FRBs).

Profile Dependence Check: The ~3 year estimate uses a logarithmic profile for simplicity. Realistic dark matter halos follow the NFW profile:

$$\rho_{NFW}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}$$

If the clock-transfer field tracks the gravitational potential ($A(\phi) - 1 \propto \Psi$), then $A(\phi) - 1 \propto \int \rho/r \, dr$, giving:

$$\alpha_{NFW}(r) \propto \ln(1 + r/r_s) - \frac{r/r_s}{1 + r/r_s}$$

For a cluster with $r_s \approx 200$ kpc and integration over $L_{halo} \approx 2$ Mpc:

- The NFW profile concentrates more delay near the core than the logarithmic profile.
- The differential delay across an Einstein radius ($r_E \approx 5$--50 kpc) is enhanced by a factor of 2–5 relative to the logarithmic estimate.
- Result: $\Delta\tilde{\tau}_{NFW} \sim 5$--$15$ years, still consistent with observed strong-lens time delays.

The order-of-magnitude estimate is robust to profile shape; realistic NFW profiles produce slightly larger delays than the toy logarithmic model.

Box 3.2: Order of Magnitude Estimate for Stochastic Shear

To estimate the magnitude of the *stochastic* shear contribution (the dynamic term $\mu_s \nabla(\Delta\tilde{\tau})$), the analysis uses the updated halo-scale delays:

- **Source Velocity:** Typical cluster transverse velocity $v_s \sim 1000$ km/s at distance $D_A \sim 1$ Gpc yields an angular proper motion $\mu_s \approx 2 \times 10^{-4}$ arcsec/year.
- **Delay Gradient:** From Box 3.1, a delay of ~ 3 years varying over arcsecond scales gives $\nabla(\Delta\tilde{\tau}) \sim 3$ years/arcsec.
- **Resulting Stochastic Shear:** The product is dimensionless shear:

$$\gamma_{stoch} \approx (2 \times 10^{-4}\,\text{arcsec/yr}) \times (3\,\text{yr/arcsec}) \approx 6 \times 10^{-4}$$

This stochastic contribution ($\gamma_{stoch} \sim 10^{-3}$) is small compared to typical weak lensing shear ($\gamma \sim 0.01\text{--}0.1$), confirming that the dynamic term is a perturbation (excess scatter), not the dominant signal. The coherent "Dark Matter" halo signature is the combined static temporal-field contribution as registered by the lens model: its physical optical-tidal component is carried by scalar backreaction on $g_{\mu\nu}$ and any bounded disformal term, while the conformal sector modifies the associated clock-transfer and inference mapping (Mechanism A). It does not arise from source motion, and not from a spatial refractive index.

3.2 Connection to Lens-Model Degeneracies

This mechanism parallels the *Source-Position Transformation (SPT)* (Schneider & Sluse 2013), which identifies a degeneracy class of mass models yielding identical observables but differing time delays. TEP extends this degeneracy into the metric sector itself. Rather than permuting the mass distribution $\kappa(\vec{\theta})$ while holding the metric constant, TEP holds the baryonic mass constant and permutes the spacetime arrival surface $\Delta\tilde{\tau}(\vec{\theta})$. Consequently, the "Phantom Mass" can be understood as a physical realization of the SPT, where the extra freedom resides in the time-transport sector rather than invisible spatial matter.

3.3 The Two Regimes: Parameter Thresholds

TEP phenomenology divides into two distinct regimes, distinguished by the magnitude of the differential clock-transfer residual and the resulting phenomenology:

Box 3.3: Regime Definitions and Parameter Thresholds

Regime	Differential clock-transfer scale	Phenomenology
Reference Envelope	ms–s	Time-domain residuals; static optical lensing effectively unchanged
Extended Regime	1--10 years	Coherent scalar-metric/disformal lensing combined with chronometric reconstruction and source-dependent Temporal-Composite response

For a 2 Mpc path, the Extended-Regime contrast is $\langle \Delta A \rangle_\gamma \sim 10^{-7}$--$10^{-6}$, consistent with the halo-scale integration in Box 3.1.

Operational Distinction: The Reference Envelope accepts the standard GW170817 translation (arrival-time offset → propagation-speed bound) at face value. The Extended Regime applies if TEP's dynamical-time interpretation is correct, in which case the standard translation may require revision—the observed $\Delta t = 1.74$ s constrains the *disformal* sector; it does not directly test the *conformal* sector, although conformal scalar sectors remain indirectly constrained by PPN, equivalence-principle, source-screening, and clock-comparison tests (Section 4).

Empirical Discriminator: The regime is determined by observation, not assumption. If lensed FRBs show only millisecond residuals, the Reference Envelope applies. If strong-lens time delays show source-dependent anomalies at the year level, the Extended Regime is indicated.

Independent Regime Criterion (Breaking Circularity): To avoid circular reasoning, the following observable specifies the regime *independently* of TEP's correctness:

- **The Variability-Mass Correlation Test:** In the Extended Regime, the inferred "dark matter" mass of a lens should correlate with the variability timescale of the background source population. Specifically: lenses observed through rapidly variable sources (AGN, quasars) should show systematically different mass reconstructions than the same lenses observed through slowly evolving sources (elliptical galaxies). This correlation is *forbidden* in standard CDM (mass is source-independent) but *required by the source-dependent Temporal-Composite component of the Extended Regime*.
- **Decision Rule:** If existing strong-lens catalogs show no statistically significant correlation between inferred lens mass and source variability class at the >3σ level, the source-dependent Temporal-Composite component is disfavored. If such a correlation exists, it constitutes positive evidence for TEP independent of FRB timing.
- **Current Status:** This test can be performed with existing data (HST strong-lens archives, SDSS quasar lenses vs. galaxy-galaxy lenses). This is flagged as a priority observational test.

Under the conservative *Reference Envelope*, $\Delta\tilde{\tau}$ is small (milliseconds to seconds). In this regime:

- **Static Lensing:** The phantom mass effect is negligible for slowly evolving sources (galaxies). Static mass maps are unaffected.
- **Time-Domain Lensing:** The effect is dominant for fast transients. A millisecond gradient across an image plane is huge for an FRB.

Conclusion: In the *Reference Envelope*, TEP is a precision correction to time-domain astrophysics, not a full dark matter substitute. In the *Extended Regime*, TEP offers a conditional geometric reinterpretation in which the component conventionally attributed to dark matter may contain an unmodeled temporal-transport contribution.

3.4 The Critical Discriminator: Variability Scatter

Since the dynamic term $\mu_{s,i} \nabla_j (\Delta\tilde{\tau})$ is randomly oriented (depending on the direction of $\vec{\mu}_s$), it does not add to the mean shear profile but contributes to the *variance* of the shear measurement.

The Variability Scatter: The inferred "shear noise" (RMS dispersion of ellipticities) should be higher for source populations with high proper motion or intrinsic variability. While the coherent "dark matter" signal is static, the *scatter* around that signal is dynamic.

This distinguishes TEP from particle dark matter, where the shear dispersion is dominated solely by intrinsic shape noise and measurement error, independent of source kinematics.

4. Reanalysis of GW170817: What is Actually Constrained?

4.1 The Measurement and the Standard Translation

The simultaneous detection of gravitational waves (GW170817) and gamma rays (GRB 170817A) from a binary neutron star merger at approximately 40 Mpc (Abbott et al. 2017) represents one of the most precise measurements in astrophysics. The signals arrived within $\Delta t_{obs} = 1.74 \pm 0.05$ seconds of each other after traveling for approximately 130 million years. This is widely cited as constraining the difference between the speed of gravity c_g and the speed of light c_γ to (e.g., Baker et al. 2017; Creminelli & Vernizzi 2017; Ezquiaga & Zumalacárregui 2017; Sakstein & Jain 2017):

> Screening in TEP is represented at the theory level by the environmental operator $\mathcal{S}_\Sigma(\mathcal{E})$. Quantities such as ρ_T, $R_T(M)$, $\mathcal{S}_\oplus(r)$, compactness Φ/c^2, local stellar density, geometric coherence length, and channel-specific response coefficients are domain-specific projections of $\mathcal{E}$, not independent screening mechanisms and not interchangeable universal thresholds. Each is an observational transfer model that parameterizes the same underlying operator in a regime-appropriate form.

$$\frac{|c_\gamma - c_g|}{c} \lesssim 10^{-15}$$

This standard translation assumes that any delay is due to a uniform difference in propagation speed accumulating linearly over the entire cosmological distance. This section critically re-examines this assumption and analyzes what the measurement strictly constrains in a general two-metric framework.

4.2 The Common-Path Invariant: A Differential Measurement

A crucial but often overlooked geometric fact is that both messengers originated from the *exact same coordinate* in distant space. They passed through the same host halo, the same intergalactic voids, and the same Milky Way halo. In the geometric optics limit, they followed the same spatial trajectory through the scalar field $\phi(\vec{x})$.

The Common-Mode Cancellation:
Because the signals originate from the same coordinate and follow a nearly identical path, they do not diverge significantly. If the metric coupling contains a conformal component $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, this factor rescales the common-mode endpoint clock and frequency calibration for *both* species identically. Any time dilation caused by passing through "time-warped regions" of space is experienced by both messengers. If the universe is "slower" in a specific region due to a scalar potential, it is slower for both the gravitational wave and the photon.

Implication for Conformal Coupling ($A(\phi)$):
As established in Section 2, a purely conformal transformation preserves null cones. *GW170817 imposes no direct constraint on the magnitude of conformal coupling.* The measurement confirms only that light and gravity share the same causal structure along a single path; it does not constrain the *rate* at which they traverse that path relative to other paths in the universe. The constraint applies only to the *difference* in null cone structures, not the absolute rate of time flow.

4.3 The Operational Reality: Decoupling the Sectors

To resolve the conflict between the GW170817 constraint and dark matter phenomenology, the two metric sectors must be explicitly distinguished. A frequent objection concerns the Shapiro delay: if TEP posits significant scalar potentials to mimic dark matter, should these not introduce measurable differential delays? The conformal sector can modify the matter-frame clock-transfer calibration associated with a Shapiro-delay measurement, while the geometric null-path delay is carried by the gravitational metric and any backreacted strong-field solution. Same-path EM–GW comparisons cancel common-mode conformal clock effects.

Since Axiom 1 establishes that photons and gravitational waves traverse the same null geodesics defined by the geometric metric, they experience an identical Shapiro delay as they propagate through the gravitational potential. Consequently, a differential arrival-time measurement like GW170817 cancels this common-mode geometric delay entirely. The conformal clock-transfer contribution is also common-mode for same-path EM–GW propagation and cancels in the differential measurement, so the magnitude of the conformal potential is not directly constrained by such differential propagation tests. It remains indirectly constrained by PPN, equivalence-principle, source-screening, gravitational-redshift, and clock-comparison tests.

Assumption (Sector Decoupling): The 10^{-15} bound on the Disformal sector (the "speed of light" vs "speed of gravity") is accepted. The TEP-GL phenomenology relies on Conformal sector gradients (the "rate of time"), which this bound does not directly constrain, and which remain bounded by their own clock-transfer, PPN, redshift, and lensing constraints after environmental screening is applied.

This decoupling makes the TEP framework robust against propagation speed constraints. Consistent with the general disformal relation (Bekenstein 1993), note that the speed of gravitational waves (c_g) constrains the causal structure (the light cone) governed by the disformal term $B(\phi)$, while the chronometric component of Phantom Mass arises from the conformal factor $A(\phi)$ (the clock rate); coherent optical lensing additionally involves scalar backreaction and any permitted disformal response. As long as $c_g = c_\gamma$, the rate at which time accumulates along that path is not determined by EM–GW arrival-time differences. GW170817 strictly constrains cone tilting, but is insensitive to the common-mode conformal time dilation that mimics mass.

Summary of Constraints:

- **Conformal Sector:** Not directly constrained by the GW170817 differential-propagation bound; independently constrained by clock, redshift, PPN, equivalence-principle, source-screening and lensing channels.
- **Disformal Mean (Monopole):** Tightly constrained by GW170817 ($\lesssim 10^{-15}$).
- **Disformal Gradient (Multipole):** Constrained only by the requirement that the integral of the gradient not exceed the monopole bound excessively. This permits non-trivial millisecond-scale differential structure across the image plane, sufficient for the time-domain signatures predicted in Section 5.

By distinguishing between the *speed of transmission* (directly constrained by GW170817) and the *rate of proper time accumulation* (constrained instead by clock, PPN, and redshift channels), the conditional phenomenological viability of TEP as a dark-sector reinterpretation is established within the screening and amplitude-closure assumptions stated here.

4.4 Environmental Screening and Solar System Constraints

A critical quantitative challenge to the TEP framework is the magnitude of the clock-transfer contrast required to affect cluster lensing reconstructions. As established in Box 3.1, halo-scale integration over $L_{\rm halo} \sim 2$ Mpc gives $L_{\rm halo}/c \simeq 6.5 \times 10^6$ yr, so a path-averaged conformal contrast $\langle \Delta A \rangle_{\rm path} \sim 10^{-7}$--$10^{-6}$ yields year-scale differential clock-transfer — commensurate with observed strong-lens delays. This replaces the earlier cosmological-integration estimate, which produced unobserved 10^3--10^5 year gaps and is not the scale required by the Extended Regime. Even at this reduced contrast, if the same field gradients persisted unmodified within the Solar System they would violate high-precision ephemerides constraints (e.g. the Cassini parameter γ).

Screening in TEP is defined at the theory level by the canonical environmental operator, which acts on the observable Temporal Shear $\Sigma_\mu \equiv \nabla_\mu \ln A(\phi)$:

$$\Sigma_\mu^{\rm obs} = \mathcal{S}_\Sigma(\mathcal{E})\, \Sigma_\mu,$$

where the environmental state $\mathcal{E}$ includes source structure, scalar gradients, density, compactness, boundary conditions, and coherence scale. These quantities determine the projection of the underlying temporal field into an observable clock, dynamical, or lensing response. TEP screening is defined directly by the environmental projection of Temporal Shear. No separate screening radius, thin-shell boundary, or imported modified-gravity mechanism is introduced.

Box 4.1: Canonical Temporal-Gradient Suppression

TEP screening acts directly on the observable Temporal Shear,

$$\Sigma_\mu \equiv \nabla_\mu \ln A(\phi),$$

through the canonical environmental operator:

$$\Sigma_\mu^{\rm obs} = \mathcal{S}_\Sigma(\mathcal{E})\, \Sigma_\mu, \qquad 0 \le \mathcal{S}_\Sigma(\mathcal{E}) \le 1.$$

The environmental state $\mathcal{E}$ includes temporal-gradient strength, density, compactness, source structure, boundary conditions, and coherence scale. These quantities determine the projection of the underlying temporal field into an observable clock, dynamical, or lensing response.

In Solar-System conditions,

$$\mathcal{S}_\Sigma(\mathcal{E}_\odot)\, |\Sigma_\mu| \ll |\Sigma_\mu|,$$

suppressing the locally observable response and preserving precision tests of GR. Across extended halo and intergalactic paths,

$$\int_\gamma \mathcal{S}_\Sigma(\mathcal{E})\, \Sigma_\mu \, dx^\mu \neq 0,$$

allowing a cumulative open-path clock-transfer contrast even when the local response is strongly suppressed. The required hierarchy is therefore a hierarchy in the environmental projection of Temporal Shear.

The saturation density ρ_T may be used as an environmental proximity indicator within $\mathcal{E}$, but it is not a hard universal threshold. The fundamental screening object remains $\mathcal{S}_\Sigma(\mathcal{E})$.

5. Observational Predictions: The Era of Chronometric Mapping

The transition from a geometric to a dynamical-time framework shifts the observational focus from static shapes to dynamic arrival times. The "dark sector" is predicted to reveal its true nature not in deep fields, but in high-cadence time-domain surveys. Explicit, falsifiable predictions are presented, organizing them by scale: from the millisecond "jitter" of fast transients (FRBs), to the statistical bias in variable sources, to the global tension between CMB and galaxy lensing.

5.1 The "Jittering" of Lensed Transients

Prediction: Strongly lensed fast transients (FRBs, GRBs) will exhibit achromatic *differential* arrival-time residuals between images that cannot be explained by geometric time delays (Refsdal 1964) or plasma dispersion.

In standard GR, the time delay Δt_{geom} between images is fixed by the mass distribution. In TEP, there is an additional arrival-time/clock-transfer residual $\Delta\tilde{\tau}$. Because ϕ fields in halos may have substructure (or "weather"), this residual varies across the image plane. For a millisecond-duration FRB (e.g., Muñoz et al. 2016), even a tiny gradient in $\Delta\tilde{\tau}$ will manifest as a timing anomaly in the *relative* arrival times of the images. Unlike plasma dispersion (which scales as ν^{-2}), this differential delay is *achromatic* (frequency-independent). The residual is expected to depend on integrated path depth and intervening temporal-field structure, and may therefore correlate with observed redshift; in TEP, redshift here is an observational clock-distance label, not a measure of physical spatial expansion. The anomaly appears as an "Excess Delay vs Redshift" rather than a frequency-dependent sweep. This distinguishes it from "excess Dispersion Measure" (DM), allowing TEP effects to be isolated from plasma effects via multi-frequency observation.

Target Candidates: Recent literature has identified specific anomalies suitable for this test. The repeating source *FRB 20190520B* exhibits a "Dispersion Measure Excess" (~ 900 pc cm^{-3}) relative to its redshift (Koch Ocker et al. 2022), currently attributed to extreme host density. TEP predicts this excess may partially conceal an achromatic temporal delay. Additionally, *FRB 20190308C*

(Chang et al. 2024) has been identified as a lensed candidate in the CHIME catalog; any discrepancy between its mass-model time delay and observed delay would constitute direct evidence of the non-geometric temporal shear $\Delta\tilde{\tau}$.

Test Protocol. No securely multiply-imaged FRB with a published independent lens model is available at the time of writing, so no lensed-FRB residual is claimed here as evidence. The prediction is instead stated as an identifiable measurement, using the blind-prediction residual of Axiom 3 (Section 2.3):

$$R_{ij} = \Delta t_{ij}^{\mathrm{obs}} - \Delta t_{ij}^{\mathrm{GR}},$$

where $\Delta t_{ij}^{\mathrm{GR}}$ must be frozen from an independent lens model before the timing comparison is performed. Each element below is required for the test to be diagnostic; a measurement missing any one of them cannot discriminate TEP from astrophysical delay structure.

Required element	Specification
Secure multiple imaging	Consistent sub-arcsecond localization and repeated burst morphology across images
Independent lens model	Frozen, pre-registered prediction $\Delta t_{ij}^{\mathrm{GR}}$ with propagated model uncertainty
Observed delay	$\Delta t_{ij}^{\mathrm{obs}}$ at high time resolution, corrected for plasma dispersion
Residual	R_{ij}, with plasma, microlensing, and host-scattering budgets subtracted (Section 2.3)
Discriminator	Achromatic residual with non-zero ensemble mean, correlated with path environment

Table 5.1: Required elements of a diagnostic lensed-FRB timing test. The achromaticity requirement separates a temporal-transport residual from plasma dispersion ($\propto \nu^{-2}$); the non-zero ensemble mean separates it from microlensing, which averages to zero with RMS scaling as $1/\sqrt{N}$.

5.2 The Variability Scatter Relation

Prediction: The *dispersion* (scatter) of weak-lensing shear measurements should correlate with the variability/kinematics of the background source population.

As derived in Section 3, the coherent "dark matter" halo signal combines scalar-backreacted optical geometry and any bounded disformal response with the conformal clock-transfer reconstruction. The Dynamic Shutter supplies the additional source-dependent component. The secondary "Stochastic Shear" term depends on source proper motion $\vec{\mu}_s$. Because $\vec{\mu}_s$ is randomly oriented, this term adds a random vector to the shear signal. TEP predicts that if one constructs a shear map using highly variable or fast-moving sources, the shear RMS will be systematically higher than for static sources, even if the mean profile (the halo) is identical.

Box 5.2: The Einstein Cross Test (Existing Data)

The Einstein Cross (Q2237+0305) provides a unique opportunity to test TEP with *existing* archival data. This quadruply-imaged quasar has:

- **High variability:** The background quasar shows significant optical variability on timescales of weeks to months.
- **Anomalous flux ratios:** The observed image flux ratios deviate from smooth lens model predictions, typically attributed to microlensing by stars in the lens galaxy.
- **Extensive monitoring:** Decades of photometric data exist (OGLE, Gaia, HST).

TEP Prediction: If the flux ratio anomalies are (partially) due to temporal shear rather than stellar microlensing, they should correlate with the quasar's variability timescale. Specifically:

- During periods of rapid quasar variability, flux ratio anomalies should be *larger*.
- During quiescent periods, anomalies should regress toward the smooth-lens prediction.
- Unlike stellar microlensing, which is uncorrelated with the source's intrinsic state, TEP predicts that "microlensing-like" anomalies will be coherent with the quasar's own variability phases—effectively turning "on" during violent source activity.

Status: This test can be performed immediately using archival OGLE light curves cross-correlated with flux ratio measurements. A positive detection would constitute strong evidence for TEP using existing data, independent of future FRB observations. This is flagged as a priority archival analysis.

5.3 The CMB-Galaxy Lensing Tension

Prediction (Extended Regime): TEP-GL predicts that galaxy weak-lensing measurements can contain a source-dependent covariance contribution absent from CMB lensing, even when their mean inferred amplitudes are statistically consistent.

While the CMB is effectively a static backlight (zero intrinsic evolution), galaxy sources are dynamic population. Under TEP, the stochastic component of the clock-transfer signal (driven by source proper motion, see Section 5.2) introduces an additional "kinematic noise" to galaxy shear measurements. Because the CMB is static, it is immune to this effect. Consequently, TEP predicts that precision cosmology inferred from galaxy weak lensing carries an unmodeled source-dependent covariance term absent from CMB lensing, which is a candidate contributor to the observed tension in the clustering amplitude (S_8).

Box 5.3: Source-Class-Dependent Shear Covariance

Observed Situation: Current weak-lensing results are survey-dependent: DES Y6 continues to favour a lower S_8 than the combined CMB determination, whereas KiDS-Legacy is statistically consistent with Planck. TEP-GL therefore does not identify a universal mean S_8 offset as its prediction; its distinctive prediction is an additional source-dependent covariance that should track source temporal structure and survey selection.

- *Planck* CMB (2018): $S_8 = 0.834 \pm 0.016$
- DES Y6 (2025): $S_8 = 0.789 \pm 0.012$, $\sim 2.6\sigma$ from combined Planck+ACT+SPT
- KiDS-Legacy (2025): $S_8 = 0.815^{+0.016}_{-0.021}$, 0.73σ from Planck

TEP Interpretation: Previous iterations of TEP considered secular source evolution as a driver for coherent shear offsets, but the magnitude ($\sim 10^{-6}$) is too small to explain the 5% tension. Instead, the tension is interpreted as a Systematic Noise Bias arising from Stochastic Shear (Box 5.2).

Mechanism: Standard maximum-likelihood estimators weight data by the inverse covariance (C^{-1}). Standard analyses assume shear noise is dominated by random galaxy orientations ("shape noise"). If the covariance matrix omits a source-dependent kinematic term (σ_μ^2), the relative weighting of high-variance regions such as cluster outskirts is misspecified, and the recovered clustering amplitude is biased.

- **Source-Dependent:** Correlated with galaxy type and redshift (via proper motion).
- **Unmodeled:** Not present in standard covariance matrices.

Result: Unmodeled excess variance in the likelihood analysis acts as a potential source of systematic bias. However, a zero-mean covariance term does not by itself determine the sign of the inferred S_8 shift; establishing the direction requires an explicit estimator-level calculation for each survey pipeline, which is not carried out here. The CMB, being static, carries no such kinematic covariance term.

Prediction: The falsifiable TEP-GL prediction is not a specific S_8 shift but the existence of an additional source-dependent shear covariance: after standard shape-noise controls, residual shear covariance should vary with source variability, proper motion, or temporal structure. The cosmological mean-growth prediction belongs to the TEP-HC (Paper 18) perturbation closure; the GL-specific discriminator is this source-class dependence. Whether it contributes to the observed S_8 tension, and in which direction, depends on the survey estimator.

5.4 Comparison of Predictions: TEP-GL vs. Particle Dark Matter

The following table summarizes the distinguishing predictions of the two frameworks:

Observable	Particle DM Prediction	TEP-GL Prediction
Lensed FRB timing	GR lens delays from baryons, particle dark matter and substructure; no TEP-specific residual correlated with temporal-field environment after lens-model controls	Achromatic *residual* anomaly (ms-scale)
Source-dependent Shear	No TEP-specific source-temporal correlation after intrinsic-alignment, selection and measurement-systematic controls	**Excess Scatter** (noise scales with μ_s)
Source variability–lens mass correlation	None (mass is source-independent)	Source-class-dependent reconstruction residual correlated with source temporal structure; amplitude determined by κ_{lens} and the environmental transfer function
CMB lensing	Standard convergence	Coherent scalar-metric lensing remains active; only the source-dependent Temporal-Composite term vanishes

Observable	Particle DM Prediction	TEP-GL Prediction
Galaxy weak lensing	Coherent component as CMB; plus source-dependent Temporal-Composite term	Additional source-class-dependent covariance absent from the CMB Temporal-Composite channel; TEP-GL does not independently fix the sign of the mean shift
Chromatic dependence	None	None (both achromatic)
Direct detection experiments	Expected signal (WIMP recoil accounting for lensing/rotation/CMB)	No signal required; TEP interprets dark sector as Phantom Mass from temporal-transport geometry, not particulate substance

A distinctive TEP-GL discriminator is the additional source-dependent component of the inferred lensing signal. The coherent scalar-metric contribution is source-independent, while the Temporal-Composite contribution is predicted to correlate with source variability, proper motion, or temporal structure.

5.5 Falsification Criteria

TEP-GL is a falsifiable hypothesis. The following observations would exclude specific regimes or the framework entirely:

1. **Reference Envelope Exclusion:** If precision timing of strongly lensed FRBs yields achromatic residuals consistent with zero to better than 0.1 ms across diverse lens environments, the Reference Envelope parameter space is excluded.
2. **Universal Shear Scatter:** If the intrinsic scatter of weak-lensing shear measurements is identical across all source kinematic classes (after correcting for measurement noise), the stochastic temporal shear mechanism is excluded.
3. **CMB-Galaxy Agreement:** CMB–galaxy agreement below 1%, after source and survey controls, excludes a Temporal-Composite contribution above that level. It does not exclude the coherent scalar-metric component of the Extended Regime.
4. **Chromatic Anomaly:** If any timing or morphological anomaly shows wavelength dependence after correction for plasma dispersion and dust, the achromatic prediction of two-metric coupling is falsified. This would indicate conventional astrophysical systematics rather than metric effects.
5. **Direct Detection:** A confirmed detection of dark matter particles that quantitatively accounts for the relevant lensing, rotation-curve, and CMB phenomenology would falsify the strong TEP dark-sector ontology. Detection of an exotic particle that does not account for these observations would not by itself falsify TEP; the particle would be an additional component, and TEP would then be tested by whether temporal-transport geometry still predicts residuals unaccounted for by the particle model.

Joint satisfaction of these null thresholds excludes the distinctive chronometric and Temporal-Composite predictions at the amplitudes specified here. The coherent scalar-backreaction sector is separately tested by CMB and galaxy lensing consistency against its frozen HC/GL prediction. The framework does not claim immunity from observation; the claim is that the observations required to test TEP have not yet been performed with sufficient precision.

Box 5.1: Quantitative Falsification Thresholds

To ensure rigorous falsifiability, explicit null-result thresholds are specified for each prediction channel:

Test	Sample Size	Precision Required	Null-Result Threshold	Regime Falsified
Lensed FRB Timing	≥10 lensed FRBs	<0.1 ms timing	95% upper bound on ensemble achromatic residual amplitude below 0.1 ms	Reference Envelope
Shear-Variability Correlation	≥1000 lenses per class	<0.5% shear scatter	No correlation at >3σ	Temporal-Composite dynamic channel
CMB-Galaxy S_8 Tension	CMB-S4 + LSST	<1% S_8 agreement	Statistically consistent agreement ($< 1\sigma$)	Temporal-Composite contribution above 1%
Variability-Mass Correlation	≥100 lenses with varied sources	10% mass precision	No correlation at >3σ	Source-dependent reconstruction channel

Joint Channel Exclusion: If all four null-result thresholds are met simultaneously, the distinctive chronometric and Temporal-Composite predictions of TEP-GL are excluded at the amplitudes specified here. The coherent scalar-backreaction sector remains independently testable through its frozen HC/GL prediction for CMB and galaxy lensing. TEP-GL as a complete lensing realization is excluded only if both the time-domain/source-dependent channels and the independently specified coherent optical prediction are ruled out.

Timeline: Tests (1) and (4) are achievable within 5–10 years (CHIME, DSA-2000, Rubin/LSST). Tests (2) and (3) require next-generation surveys (CMB-S4, Euclid) with ~10-year horizons. A definitive verdict on TEP-GL is therefore expected by ~2035.

6. Discussion: A Paradigm Shift in the Dark Sector

6.1 Limitations of the "Stack of Corrections"

Gravitational physics has historically advanced by refining the fundamental geometric framework rather than adding ad-hoc corrections. TEP represents a similar upgrade to the fundamental object, following the natural historical progression of physical theory:

- **Newton:** Gravity is a Force; Time is Absolute.
- **Einstein:** Gravity is Geometry; Time is Relative (Coordinate-Dependent).
- **TEP:** Gravity is Geometry; Time is a *Dynamical Field*.

By treating the rate of proper time accumulation as a physical field with its own degrees of freedom (rather than a fixed function of the metric), TEP unifies the "stack" into a single framework. The "dark matter" anomaly is the observation that this field has spatial gradients.

6.2 CMB Lensing and the Integrated Sachs-Wolfe Constraint

A scalar field capable of generating year-scale differential delays on cluster scales raises a critical question: what are the implications for the Cosmic Microwave Background (CMB)? Two potential tensions must be addressed.

6.2.1 The ISW Effect

The Integrated Sachs-Wolfe (ISW) effect arises when CMB photons traverse time-evolving gravitational potentials. In TEP, the conformal factor $A(\phi)$ contributes an additional term to the photon temperature perturbation:

$$\frac{\Delta T}{T}\bigg|_{\text{TEP}} = \int \frac{\partial \ln A(\phi)}{\partial t}\, dt$$

For TEP to remain consistent with *Planck* constraints on the ISW amplitude, one of the following must hold:

- **Quasi-Static Regime:** The scalar field evolves slowly ($\partial_t A \ll H_0 A$), so the TEP contribution to ISW is subdominant to the standard $\dot{\Psi}$ term. This is natural if ϕ tracks the matter distribution adiabatically.

- **Cancellation:** In some scalar-tensor theories, the conformal ISW contribution partially cancels the metric ISW, leaving the net effect within observational bounds (cf. Amendola et al. 2008).

Box 6.1: Order-of-Magnitude ISW Consistency Check

Conditional on the HC tracking closure in which ϕ follows the matter distribution adiabatically, the resulting order-of-magnitude estimate satisfies the quasi-static ISW bound.

Setup: The conformal factor is $A(\phi) = 1 + \alpha(\phi)$ with $\alpha \sim 10^{-6}$ on halo scales. The following estimate uses the conventional reference-background variables employed by the HC linear-transfer representation (Paper 18). Here $H_{\rm ref}(z)$ is an observational/reference inverse-time scale and $a_{\rm eff} = A_{\rm clock}$; neither denotes physical expansion of the underlying spatial geometry. The ISW constraint requires:

$$\left|\frac{\partial \ln A}{\partial t}\right| \approx \left|\frac{\dot{\alpha}}{1+\alpha}\right| \approx |\dot{\alpha}| \ll H_0$$

Estimate of $\dot{\alpha}$: If ϕ tracks the matter distribution adiabatically (sourced by T^{μ}_{μ}), then α evolves on the timescale of structure formation:

$$\dot{\alpha} \sim \alpha \cdot \frac{\dot{\rho}}{\rho} \sim \alpha \cdot H_{\rm ref}(z) \cdot f_{\rm ref}(z)$$

where $f_{\rm ref}(z) \equiv d\ln D/d\ln a_{\rm eff} \approx \Omega_m(z)^{0.55}$ is the growth rate and $a_{\rm eff} = A_{\rm clock}$. At $z \sim 0.5$ (peak ISW sensitivity), $f_{\rm ref} \approx 0.8$ and $H_{\rm ref}(z) \approx 1.2 H_0$.

Result:

$$\frac{|\dot{\alpha}|}{H_0} \sim \alpha \cdot f_{\rm ref} \cdot \frac{H_{\rm ref}(z)}{H_0} \sim 10^{-6} \times 0.8 \times 1.2 \approx 10^{-6}$$

This is six orders of magnitude below the ISW constraint threshold ($\lesssim 0.1$). The quasi-static approximation is therefore satisfied under this tracking closure; the result should not be interpreted as independent derivation of that closure within TEP-GL.

Physical Interpretation: The scalar field ϕ varies spatially (creating "dark matter" halos) but evolves temporally only as fast as the underlying matter distribution. Year-scale *spatial* delays across an image plane are compatible with cosmologically slow *temporal* evolution because the delay gradient is set by the static $\nabla\alpha$, not by $\dot{\alpha}$.

6.2.2 CMB Lensing Power Spectrum

The CMB lensing convergence power spectrum $C_\ell^{\kappa\kappa}$ is measured to high precision by *Planck* and ACT. In standard ΛCDM, this is sourced entirely by the integrated mass distribution. Consistent with the optical-tidal decomposition of §3.1.4, the physical angular remapping in TEP is carried by the baryonic term, the scalar-backreaction correction to the Einstein-frame geometry, and any permitted disformal contribution:

$$\kappa_{\rm TEP}^{\rm phys} = \kappa_{\rm baryon} + \delta\kappa_{g[\phi]} + \delta\kappa_B$$

The conformal sector may additionally enter the *inferred* convergence through the chronometric reconstruction mapping,

$$\kappa_{\rm TEP}^{\rm inferred} = \kappa_{\rm TEP}^{\rm phys} + \delta\kappa_{A,\,\rm recon},$$

but it does not independently deflect the CMB rays, since conformal transformations preserve null cones (Axiom 1).

For the CMB (a static, non-evolving source), the Temporal-Composite Shear term vanishes identically—there is no source proper motion or intrinsic variability on lensing timescales to couple to the delay gradient. Removing that term does *not* make the CMB insensitive to TEP: it means CMB lensing constrains the coherent scalar-metric perturbation sector rather than source-evolution smearing.

Implication: CMB lensing therefore tests the coherent scalar-metric sector, which is implemented in TEP-HC (Paper 18) through the pure-conformal Bellini–Sawicki closure with active $\delta\phi$ evolution. TEP-GL imports that linear closure rather than treating CMB lensing as a source-evolution effect. The amplitude is controlled by the integrated scalar-backreaction profile, not by lens motion; consistency with the observed spectrum constrains the relationship between $\alpha(\phi)$ and the matter density ρ.

6.2.3 Parameter Space Constraints

Combining ISW and CMB lensing constraints with the continuous gradient-suppression screening framework (Box 4.1) defines a narrow but viable parameter window:

Constraint	Requirement	Status
Solar System (Cassini)	$\mathcal{S}_\Sigma(\mathcal{E}_\odot)\,\lvert\Sigma\rvert \ll \lvert\Sigma\rvert$	Satisfied via environmental suppression
Cluster Lensing	$\left\lvert\int_{\gamma_{\rm halo}} \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu\,dx^\mu\right\rvert \sim 10^{-7}\text{--}10^{-6}$	Required for DM phenomenology; value set by solved transfer function
ISW (*Planck*)	$\partial_t \ln A/H_0 \lesssim 0.1$	Conditional on HC tracking closure
CMB Lensing	$\delta\kappa_{g[\phi]} + \delta\kappa_B$ must satisfy the HC/CMB lensing spectrum constraint; $\delta\kappa_{A,\rm recon}$ treated separately	Constrains the coherent scalar-metric closure and its relationship to the matter distribution

The required closure is a relationship between the solved scalar-backreaction profile and the matter distribution. It is not a claim that the pure conformal factor independently deflects CMB rays. The closure is satisfied in models where the scalar-backreaction profile is sourced by the matter density, but represents a theoretical prior that must be verified against N-body simulations. This is flagged as a key target for future numerical work.

Box 6.3: Cross-Scale Temporal-Gradient Constraint

A viable TEP realization must produce a strongly suppressed observable Temporal Shear in local high-precision environments while retaining a nonzero integrated temporal-gradient response across extended halo paths:

$$\mathcal{S}_\Sigma(\mathcal{E}_\odot)\,\lvert\Sigma\rvert \le \Sigma_{\rm local}^{\rm max},$$

$$\left\lvert\int_{\gamma_{\rm halo}} \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu\,dx^\mu\right\rvert \sim 10^{-7}\text{--}10^{-6}.$$

These are constraints on one continuous environmental operator, not independently adjusted couplings at different scales. The Solar-System bound is satisfied when the environmental projection $\mathcal{S}_\Sigma(\mathcal{E}_\odot)$ suppresses the locally observable Temporal Shear below precision-test thresholds. The halo constraint is satisfied when the same operator, acting across an extended path through lower-density environments, permits a cumulative clock-transfer contrast of order $10^{-7}\text{--}10^{-6}$.

6.3 Interpretational Challenges

TEP effects mimic standard lensing signatures, making them difficult to distinguish without time-domain analysis.

1. **Static Degeneracy:** Most lensing data are analyzed under static assumptions. In this limit, temporal shear is mathematically indistinguishable from spatial mass (Section 3).
2. **Differential vs. Common-Mode Effects:** The GW170817 constraint is often interpreted as excluding modified metric couplings. However, as shown in Section 4, this constraint applies to the differential disformal sector. Conformal proper-time dilation is a *common-mode effect* for co-propagating signals, leaving the "speed of gravity" constraint satisfied while permitting significant time-domain phenomenology.

6.4 Reinterpreting the Evidence: Multipath vs. Single-Path

When viewed through the lens of TEP, the disparate constraints on the dark sector resolve into a consistent picture based on signal topology:

- **Lensing (Multipath):** Observations that require "dark matter" (e.g., lensing) are fundamentally *multipath* measurements. These different pathways sample different values of the time field ϕ, creating differential delays that manifest as "phantom mass."
- **GW170817 (Single-Path):** Observations that constrain "modified gravity" (GW170817) are fundamentally *single-path* measurements. The signals originate from the same coordinate and traverse the same time-warped regions. The temporal distortions are common-mode and cancel out.

The apparent contradiction—"how can gravity be modified enough to create dark matter but unmodified enough to pass GW170817?"—is resolved. TEP introduces the temporal field as the additional physical degree of freedom; its conformal sector governs clock transport without bending null geodesics, while its stress-energy and strong-field operators can backreact on the Einstein-frame geometry (Bahrain, Paper 28). The differential-propagation bound constrains the disformal sector, not the common-mode conformal clock sector. Time manifests as a differential observable only when comparing divergent paths.

6.5 Path-Dependent Distance/Proper-Time Ratio

If TEP is correct, then local c remains invariant, but the *inferred ratio between spatial separation and matter-clock transfer registered between endpoints* along extended paths is path dependent. This challenges the foundational assumption used to interpret all astrophysical data.

- **Standard Assumption:** The universal clock rate is constant; therefore, any anomaly in t must be due to extra path length (spatial curvature) or extra mass (Shapiro delay). Conclusion: particulate *dark matter* is inferred.
- **TEP Reality:** The matter proper-time accumulation rate $d\tilde{\tau} = A(\phi)d\tau_g$ varies with location. The anomaly in inferred travel time is due to the scalar field gradient modulating clock rates along the path, not to a variation in local c. Conclusion: apparent *dark matter* is reconstructed temporal transport.

TEP removes the need for an invisible substance by correcting the assumption that a single universal clock rate governs all paths, rather than by asserting a variable speed of light.

This is the local multipath analogue of the cosmological TEP mapping: conventional cosmology reconstructs observed redshift as spatial expansion, whereas TEP attributes it to the evolution of the conformal proper-time map on an underlying static spatial geometry (Athens, Paper 26; Thika, Paper 27). In lensing, the corresponding reconstruction can project temporal transport into inferred spatial mass. The two effects — dark matter and cosmic expansion — are parallel temporal-to-spatial reconstruction artefacts in TEP, not independent ad hoc mechanisms.

6.6 TEP as a Bridge Between Paradigms

TEP offers a potential unification of competing frameworks:

- **CDM:** In the limit where the scalar field ϕ relaxes into stable halos and source variability is ignored, TEP reproduces the phenomenology of Cold Dark Matter (via the Phantom Mass mechanism).
- **MOND:** In low-acceleration environments, environmentally active Temporal Shear supplies a candidate common origin for MOND-like scaling. This paper establishes the acceleration-scale correspondence but does not independently derive the complete MOND interpolation law.

TEP provides a single temporal-gradient framework within which CDM-like large-scale phenomenology and MOND-like low-acceleration behaviour may emerge in different environmental projections.

Box 6.2: Relation to Low-Acceleration Phenomenology

In TEP, MOND-like phenomenology is associated with the environmental activation of observable Temporal Shear in low-acceleration, weakly suppressed systems. The canonical screening operator $\mathcal{S}_\Sigma(\mathcal{E})$ becomes weak in low-gradient galactic environments, allowing the observable Temporal Shear response to become active. This supplies a candidate origin for MOND-like phenomenology without modifying the local value of c.

The empirical coincidence

$$a_T \sim cH_0$$

identifies the relevant observational acceleration scale, where H_0 is the conventional inverse-time scale encoded by the cosmological clock map rather than physical spatial expansion. In TEP, H_0 is not identified with $\dot{a}_m/a_m$, since $a_m = 1$.

Numerical Check:

$$a_0 \sim cH_0 \approx (3 \times 10^8 \text{ m/s}) \times (2.2 \times 10^{-18} \text{ s}^{-1}) \approx 6.6 \times 10^{-10} \text{ m/s}^2$$

This lies within a factor of about five of the empirical MOND value $a_0 \approx 1.2 \times 10^{-10}$ m/s². TEP attributes this cross-scale relation to the same temporal-gradient structure represented by $\mathcal{S}_\Sigma(\mathcal{E})$. The defensible statement is that TEP produces an acceleration scale of cosmological magnitude, which suggests that TEP and MOND phenomenology may share a common origin. This matches the level of claim established in TEP-UCD (Paper 6); TEP-GL does not independently derive the numerical MOND acceleration scale.

Physical Interpretation: Low-acceleration phenomenology arises in TEP because the environmental operator $\mathcal{S}_\Sigma(\mathcal{E})$ leaves the Temporal Shear response active precisely in low-gradient regions. Regions with internal accelerations $a \lesssim cH_0$ are weakly screened and the temporal gradient contributes; regions with $a \gtrsim cH_0$ are strongly screened and follow Newtonian dynamics. The low-gradient scalar profile, its metric backreaction, and the reconstruction response must be fitted jointly to galaxy and lensing data before the MOND transition can be claimed as derived.

6.7 Convergence of Evidence

While TEP-GL is a theoretical proposal, it is notable that multiple independent anomalies in current cosmology converge on the phenomenology predicted by this framework. These tensions, often treated as separate puzzles, may represent a single systemic failure of the isochrony assumption.

TEP-GL Prediction	Existing Observational Anomaly	Status
Source-Dependent Shear (Kinematic noise bias)	**S_8 Tension** Survey-dependent: DES Y6 lower than CMB; KiDS-Legacy consistent with Planck.	**Motivating Consistency**
Mass-Sheet Degeneracy (Temporal vs Spatial)	**H_0 Tension** Time-delay cosmography ($H_0 \approx 73$) conflicts with CMB ($H_0 \approx 67$).	**Consistent**
Temporal-Composite Shear (Dynamic Shutter)	**Flux Ratio Anomalies** Substructure required to explain ratios is often not found; "phantom" substructure.	**Consistent**
Variability-Shear Correlation	**Einstein Cross Anomalies** Existing Einstein Cross monitoring provides archival data suitable for testing the predicted variability-phase correlation; no dedicated TEP correlation analysis has yet been performed (Eigenbrod et al. 2008).	**Untested**

The theory does not require new observations to find initial support; it provides a unified explanation for *existing* anomalies that Λ CDM struggles to explain simultaneously.

6.8 The Path Forward

The analysis suggests that *dark-matter phenomenology may be reproducible without a new particulate dark component*, if the combined coherent optical, chronometric and Temporal-Composite predictions survive the proposed tests. In the TEP ontology, the underlying entity is a temporal field whose geometry and transport are reconstructed as apparent spatial mass under the Isochrony Axiom. For particulate dark matter to remain the complete explanation of the relevant anomalies, the distinctive TEP residuals must be excluded at the required precision. The framework also provides a common setting in which CDM-like and MOND-like phenomenology may emerge, while the quantitative MOND transition remains to be derived.

"Dark Time" is not merely an alternative label for dark matter; within the Extended Regime, it offers a geometric identification of the unmodeled temporal-transport contribution that may be embedded within the phenomenology conventionally attributed to dark matter.

7. Conclusions

7.1 Limitations of the Isochrony Axiom

For nearly a century, dark-matter inference has been developed within a framework that treats the residual source-time structure relevant to static reconstruction as effectively synchronous once standard propagation delays are modeled. By treating the photons in telescopes as representing a single spatial slice of the source, standard models have been forced to interpret all observed distortions as spatial deflections caused by mass. It has been shown that this *Isochrony Axiom* is an effective approximation that may break down in the presence of generalized metric couplings. In the TEP ontology, the dark sector is not fundamental matter or energy; it is part of the residual produced when dynamical temporal transport is forced into a synchronous spatial reconstruction. The same interpretive principle applies cosmologically: observed redshift is reconstructed conventionally as physical spatial expansion, whereas TEP attributes it to the conformal proper-time map on an underlying static spatial geometry (Athens, Paper 26; Thika, Paper 27). Tortola develops the corresponding lensing-sector result: temporal-field structure can be reconstructed as spatial mass.

Under TEP, the dark lensing signal is the combined projection of three manifestations of one temporal field: coherent optical geometry generated through scalar backreaction on $g_{\mu\nu}[\phi]$ and any permitted disformal response; conformal open-path chronometric reconstruction through $A(\phi)$; and source-dependent Temporal-Composite response. When these contributions are reconstructed under an isochronous GR model, they appear as additional spatial mass. In the TEP ontology, this Phantom Mass is real temporal-field phenomenology, but it is not particulate dark matter.

7.2 Summary of Findings

1. **Phantom Mass from Dark Time:** TEP-GL demonstrates how one temporal field projects into coherent optical, chronometric-reconstruction and source-dependent image channels. Scalar backreaction and any permitted disformal contribution generate coherent angular response; conformal open-path clock transfer changes timing and inferred mass normalization; source evolution generates Temporal-Composite Shear. Standard isochronous reconstruction combines these effects into an apparent dark-matter distribution.
2. **The GW170817 "Speed Limit" is Nuanced:** The standard multi-messenger constraint was deconstructed. Because the conformal component of the metric coupling preserves null cones, it is *invisible* to differential speed-of-gravity tests. Photons and gravitational waves share the common-mode dilation. While GW170817 constrains disformal propagation speeds to $|c_\gamma - c_g|/c \lesssim 10^{-15}$, it does not directly constrain common-mode conformal clock-rate structure along the shared path, although conformal scalar sectors remain indirectly constrained by PPN, equivalence-principle, source-screening, and clock-comparison tests. The analysis demonstrates that conformal gradients may reproduce specific timing-sensitive aspects of dark-matter-like phenomenology—particularly timing-sensitive signatures and, through scalar backreaction on $g_{\mu\nu}[\phi]$ and any permitted disformal response, coherent lensing shear—subject to continuous gradient-suppression screening that ensures consistency with Solar System tests.
3. **A New Observational Era:** Two regimes for testing TEP are defined. In the conservative *Reference Envelope*, the effects are millisecond-scale and detectable only in high-time-resolution astrophysics (FRBs, pulsars). In the *Extended Regime*, where environmental screening and/or a revised operational mapping allow larger effective delays, TEP becomes a full temporal-geometry replacement for particle dark matter.

7.3 Future Outlook: Time, Not Mass

The persistence of the dark matter problem despite decades of particle searches suggests a potential error in the underlying premises. The error may lie in the treatment of time. By treating time as a passive parameter rather than an active dynamical field, there is a risk of blinding analysis to the true nature of the "dark" universe.

The path forward lies in *Chronometric Lensing*. The field must move beyond static mass-mapping and towards *chronometric mapping*—measuring the detailed arrival-time structure of the universe. If "dark matter" halos are actually "time dilation" halos, the definitive indicator will not be a particle recoil in a xenon tank, but a millisecond residual in a Fast Radio Burst. The focus of inquiry must shift from searching for missing mass to characterizing the dynamical structure of time.

7.4 The Paradigm Shift

Within the operational axioms adopted here, the phenomenology traditionally attributed to a particulate dark sector can be reinterpreted as the projection of two-metric time transport onto inference pipelines that assume isochrony. The anomalies are real; the claim is that their standard "missing mass" interpretation is not unique once the Isochrony Axiom is relaxed.

The dark sector is thus reinterpreted not as an invisible substance, but as the shadow of temporal transport.

The observational program outlined herein—time-domain lensing of fast transients, precision strong-lens residual timing, and source-variability-dependent shear consistency tests—provides a viable route to distinguish particle dark matter from a temporal-composite channel while maintaining a conservative Reference Envelope baseline anchored to existing multi-messenger constraints, thereby offering a promising avenue for resolving the dark matter problem.

Updated Operational Interpretation

The later dedicated strong-lensing analysis (TEP-LENS, Paper 19) refines the observational implementation used here. In particular, ordinary image-arrival delays remain algebraically integrable; the strong-lensing test is formulated as an observed-versus-GR-predicted transport residual rather than a literal closure violation. The framework presented here is consistent with that refinement: Axiom 3 already distinguishes open-path differential residuals (the GL observable) from closed-loop holonomy, and the static clock-transfer contribution is identified as a clock-transfer discrepancy rather than a direct refraction of null geodesics. Readers should consult TEP-LENS for the current canonical strong-lensing time-delay formulation.

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Version: v0.7 (Tortola) · First published: 19 December 2025 · Last updated: 10 August 2026

Data Availability & Reproducibility

This work follows open-science practices. This is a theoretical framework paper developing the TEP-GL (Temporal Equivalence Principle - Gravitational Lensing) model. The analysis builds upon established dark matter phenomenology from the literature and derives new observational predictions from first principles.

Repository & Code

GitHub Repository: github.com/matthewsmawfield/TEP-GL

The repository contains the complete theoretical framework, mathematical derivations, and observational predictions for the TEP-GL model.

Repository Structure

```
TEP-GL/
├── manuscripts/               # Markdown manuscript sources
│   └── manuscript-tep-gl.md
├── site/
│   ├── components/            # HTML manuscript sections
│   │   ├── abstract.html
│   │   ├── section_1_introduction.html
│   │   ├── section_2_theoretical_framework.html
│   │   ├── section_3_quantitative_comparison.html
│   │   ├── section_4_gw170817.html
│   │   ├── section_5_observational_predictions.html
│   │   ├── section_6_discussion.html
│   │   ├── section_7_conclusions.html
│   │   ├── acknowledgments.html
│   │   ├── references.html
│   │   └── appendix_a.html
│   └── manifest.json
├── scripts/                   # Analysis scripts (future expansion)
│   └── utils/
├── requirements.txt           # Python dependencies
├── CITATION.cff               # Citation metadata
└── README.md                  # Repository documentation
```

Data Provenance

This is a theoretical framework paper. Primary data references:

- **GW170817:** LIGO/Virgo gravitational wave and EM counterpart data (public)
- **Dark matter phenomenology:** Established lensing results from the literature (fully cited)
- **SPARC database:** For TEP-UCD cross-paper validation

Reproduction Instructions

Quick Start (Manuscript Build)

```
# 1. Clone repository
git clone https://github.com/matthewsmawfield/TEP-GL.git
cd TEP-GL

# 2. Build manuscript website
cd site
npm install
npm run build

# 3. Output in site/dist/
```

System Requirements

- **Node.js** 16+ (for site building)
- **Storage:** < 50 MB

Theoretical Framework Documentation

The TEP-GL framework derives from three foundational postulates:

1. **Two-Metric Postulate:** Gravitational dynamics and universal causal matter coupling are distinguished by $g_{\mu\nu}$ and $\tilde{g}_{\mu\nu}$.
2. **Temporal Transport:** Conformal structure produces open-path clock/frequency transport, while residual closed-loop synchronization non-integrability requires a disformal or otherwise non-exact sector.
3. **Chronometric Reconstruction:** Temporal-field transport and scalar-induced geometry can project into conventionally inferred spatial mass.

Software Versions

- **Node.js** 16+
- **Python** 3.8+ (optional utilities)

Appendix A: Mathematical Derivations

A.1 Proof of Null Cone Invariance in the Conformal Limit

Proposition: If two metrics are conformally related by $\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$, they share the same null geodesics as unparameterized curves.

Proof:
Let $k^\mu = dx^\mu/d\lambda$ be a null vector in $g_{\mu\nu}$, satisfying $g_{\mu\nu}k^\mu k^\nu = 0$ and the geodesic equation $k^\nu \nabla_\nu k^\mu = 0$ (where ∇ is the Levi-Civita connection of g).
In the metric $\tilde{g}_{\mu\nu}$, the null condition holds immediately:

$$\tilde{g}_{\mu\nu}k^\mu k^\nu = A^2(\phi)g_{\mu\nu}k^\mu k^\nu = 0$$

The connection coefficients $\tilde{\Gamma}^\lambda_{\mu\nu}$ for $\tilde{g}$ are related to $\Gamma^\lambda_{\mu\nu}$ by:

$$\tilde{\Gamma}^\lambda_{\mu\nu} = \Gamma^\lambda_{\mu\nu} + \delta^\lambda_\mu \partial_\nu \ln A(\phi) + \delta^\lambda_\nu \partial_\mu \ln A(\phi) - g_{\mu\nu}g^{\lambda\sigma}\partial_\sigma \ln A(\phi)$$

Substituting this into the geodesic equation for $\tilde{g}$:

$$k^\nu \tilde{\nabla}_\nu k^\mu = k^\nu \nabla_\nu k^\mu + 2k^\mu k^\nu \partial_\nu \ln A(\phi) - g_{\nu\sigma}k^\nu k^\sigma \dots$$

Since k is null ($g_{\nu\sigma}k^\nu k^\sigma = 0$) and $k^\nu \nabla_\nu k^\mu = 0$, this yields:

$$k^\nu \tilde{\nabla}_\nu k^\mu = 2(k^\nu \partial_\nu \ln A(\phi))k^\mu$$

This is the geodesic equation with a non-affine parameterization ($Dk/d\lambda \propto k$). Thus, the curve is a geodesic of $\tilde{g}$, differing only by the parameterization (the clock rate). □